Are Transformers Useful for Inflation Forecasting?

A Comparative Analysis of Transformer Models for Global Inflation

Er transformer-modeller nyttige til forudsigelse af inflation?
En sammenlignende analyse af transformer-modeller for global inflation

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Abstract

Forecasting inflation is a longstanding challenge in economics, where even simple statistical models often perform surprisingly well. Recent advances in forecasting specific transformer architectures make bold claims, yet their usefulness is still not widely accepted, and their value for inflation forecasting remains untested.

This thesis provides one of the first evaluations of transformers for inflation forecasting. Using monthly inflation data for 91 countries from 1980 to 2019, three transformer variants Informer, Autoformer, and FEDformer are compared against strong benchmarks, including random forests and the recently proposed linear neural networks DLinear and NLinear. Forecasts are evaluated across horizons using root mean squared error (RMSE), median absolute deviation (MAD), and the average superior predictive ability (aSPA) test.

The results show that transformers do not outperform the simpler models. Recursive linear neural networks, particularly NLinear, consistently deliver the lowest forecast errors, while random forests perform strongly across many country–horizon pairs. Informer is somewhat competitive at short horizons, but Autoformer and FEDformer lag behind.

Overall, the findings highlight a classic result: added complexity does not automatically improve forecasting performance. Recursive linear models and random forests remain robust and efficient tools for global inflation forecasting, while transformer-based models offer little beyond higher computational cost.

All code used in this thesis is publicly available at `https://github.com/OleSahlholdt/Inflation_master_thesis`

1 Introduction

Accurately forecasting inflation is crucial for economic stability, policy decisions, and financial planning. Inflation is the rate at which the general level of prices for goods and services rises, which has a direct impact on purchasing power, interest rates, and overall economic growth. Misestimating inflation can lead to severe consequences for governments, businesses, and individuals.

For central banks, such as the European Central Bank, precise inflation predictions are essential for setting monetary policy. If inflation is underestimated, interest rates may remain too low, fueling excessive borrowing and potentially overheating the economy. Conversely, overestimating inflation could lead to unnecessarily high interest rates, restricting economic growth and increasing unemployment.

For businesses, accurate inflation forecasts help with pricing strategies, wage negotiations, and investment decisions. Companies that fail to anticipate inflation correctly may struggle with cost management, supply chain disruptions, or reduced consumer demand. Investors and financial institutions also rely on inflation projections to assess risks in bond markets, equities, and real estate.

On an individual level, inflation affects wages, savings, and purchasing power. If inflation rises unexpectedly, fixed incomes may lose value, and savings may erode, making long-term financial planning more difficult. Consumers also adjust their spending behavior based on inflation expectations, influencing overall demand in the economy.

Despite its importance, inflation has proven to be a difficult economic variable to forecast. Unlike output or unemployment, inflation is shaped by a wide range of domestic and international forces, including commodity price shocks, supply chain disruptions, and policy interventions — that are inherently hard to predict. [16, 17]

Central banks and international organizations frequently fail to anticipate sharp turning points, such as sudden accelerations or decelerations in inflation. Forecasts are often biased, either systematically underestimating or overestimating inflation, and they tend to adjust only slowly when conditions change [20]. A striking feature of the literature is that, despite the sophistication of many proposed models, very simple benchmarks — such as autoregressive processes or even random walk models — are remarkably difficult to outperform consistently [5, 42]. This puzzle has motivated much of the inflation forecasting research.

Another layer of complexity arises from the role of subjective forecasts. Survey-based forecasts from professional forecasters or households often perform better than model-based forecasts, especially when it comes to establishing boundary conditions: a good assessment of where inflation stands today (the nowcast) and what the medium-run "ocal mean inflation rate is likely to be [20]. The fact that human judgment sometimes beats formal models raises important questions about the limits of purely model-driven forecasting and highlights the need to integrate better boundary information into model-based approaches.

In short, inflation forecasting presents both a policy necessity and a persistent intellectual challenge. The next step is to understand how economists have attempted to address this challenge, beginning with traditional macroeconomic models and moving toward modern machine learning approaches.

Given the importance of reliable inflation forecasts, models that can systematically improve prediction accuracy are highly sought for. Early approaches often relied on structural relationships grounded in economic theory, most notably the Phillips curve, which posits a negative correlation between unemployment and inflation. While influential for decades, the Phillips curve framework ultimately proved unreliable in practice. Atkeson, Ohanian, et al. (2001) showed that simple Phillips curve-based models produced forecast errors that were often larger than those from naive benchmarks, such as a random walk. Subsequent studies reinforced this view, suggesting that the information content of the Phillips curve for forecasting is weak and unstable over time [19].

Thus despite the complexity of many forecasting models, very simple benchmarks such as AR processes or random walks often perform just as well, if not better. Stock and Watson (1999), Stock and Watson (2010) emphasized that across a wide range of specifications and datasets, outperforming these simple models in a systematic way has proven difficult. For this reason, AR benchmarks have become the de facto yardstick against which new forecasting methods are judged.

The inability of traditional structural models to consistently improve forecasts reflects deeper challenges. Inflation dynamics are influenced not only by domestic labor market conditions but also by global shocks, supply disruptions, and policy interventions—forces that are difficult to capture in simple theoretical relationships [5, 42, 19]. At the same time, purely statistical models, while robust in their simplicity, often lack explanatory power and can fail during periods of structural change.

This tension between theory-driven models and statistical benchmarks has shaped much of the debate in early inflation forecasting. It also set the stage for the exploration of alternative approaches, particularly as the availability of large datasets and computational tools began to expand.

As datasets have gotten larger and computational power has increased, the emphasis on machine learning models has grown. Early research often focused on regularized linear methods, such as factor models or penalized regressions, which can handle a large number of predictors with more flexibility than more classic methods [41, 6].

More recently non-linear ML methods have shown good results, including tree-based algorithms and neural networks. Medeiros et al. (2021) provide a thorough investigation that shows that machine learning can improve inflation forecasting. Using the FredMD[30], they find that decision tree based models outperform autoregressive benchmarks, where especially random forest (RF) showed a large improvement. In follow-up work, Medeiros, Schütte, and Soussi (2023) show that these results hold across multiple countries, establishing random forests as a robust performer in inflation forecasting.

Neural networks have also been applied to this problem, with promising results even in relatively early studies [14, 34]. But with more advanced architectures, such as recurrent neural networks (RNNs) and long short-term memory (LSTM) models, that are specifically designed for sequential input. Almosova and Andresen (2023) evaluate the performance of LSTMs on the FredMD dataset and find that these models can outperform random walk forecasts and compete with traditional autoregressive and seasonal autoregressive models. This suggest, once again, that nonlinear models, offer clear advantages in capturing inflation dynamics. However, they also note limitations: in univariate settings, the performance of LSTMs is not always superior to well-specified seasonal models, and careful tuning of hyperparameters is essential.

Further evidence of the potential of deep learning comes from Barkan et al. (2023), who investigate hierarchical recurrent neural networks. They find that such models outperform several popular alternatives, including random forests and gradient boosted trees, though their focus is not on raw inflation but disaggregated inflation components of the Consumer Price Index.

In general there is a trend showing that machine learning methods—particularly tree-based algorithms and neural networks—can improve forecast accuracy compared to traditional benchmarks. Yet, results remain uneven, and no single method has emerged as clearly dominant across datasets and forecast horizons.

The rise of deep learning methods have opened new possibilities for time series forecasting, and inflation forecasting is no exception. Neural networks such as LSTMs have already shown that non-linear, sequential architectures can provide gains over both traditional econometric models and tree-based machine learning methods. However, these models still face important limitations. They can be difficult to estimate, require extensive tuning of hyperparameters, and their performance is not always stable across datasets and forecast horizons [2, 8].

In recent years, transformer architectures have received a lot of traction (likely because of huge text-based models) in the broader machine learning literature. Originally introduced for natural language processing, transformers rely on attention mechanisms that allow the model to learn long-range dependencies in sequential data without the need for recurrence as it is for recurrent neural networks. This feature has proved highly effective in a wide range of tasks outside of economics, and has motivated the adaptation of transformer-based models to time series problems. [45, 46, 48, 49].

time series forecasting using deep learning has been dominated by recurrent neural networks (RNNs) and their variants (such as LSTMs). These models process data sequentially, which constrains parallel computation and makes it difficult to learn very long-range dependencies [27]. In fact, Vaswani et al. (2017) point out that in RNNs or convolutional models the number of operations required to relate two distant positions grows with their distance, making it hard to capture far-apart relationships. The Transformer aims to overcome this by replacing recurrence with a self-attention mechanism: every pair of positions is connected in one attention step, yielding a constant "path length" between any two inputs. As a result, the Transformer allows for significantly more parallelization and makes it easier to model long-term context. Since its introduction in 2017, the Transformer has achieved state-of-the-art results in natural language processing, and has recently begun to be applied to financial time series [27].

As mentioned, there is evidence that the use of machine learning methods can enhance the forecasting of financial time series data. Now it is natural to ask whether transformers can further improve the accuracy of inflation forecasts. However, research on the application of transformers in time series forecasting is still relatively limited. A sceptical analysis by Zeng et al. (2023) raises concerns about the self-attention mechanism, arguing that it may disrupt temporal information. While modifications to transformer architectures can mitigate this issue, it cannot be entirely avoided. Zeng et al. (2023) show that these transformer-based models are not performing better than simple linear neural networks. And argues that the benefit of the transformer models only comes from the ability to forecast several steps into the future without relying on recursion.

Although Zeng et al. (2023) question the usefulness of transformers in time series forecasting, several recent studies demonstrate that transformer-based architectures can overcome the shortcomings of recurrent models and achieve state-of-the-art results on financial and economic datasets [48, 46, 49]. Given the importance of accurate inflation forecasts and the relative lack of transformer applications in this context, it is timely and necessary to evaluate whether these models can provide improvements in inflation forecasting.

2 Research Gap and Contribution

The literature on inflation forecasting has traditionally emphasized simple econometric models and, more recently, a variety of machine learning approaches. While tree-based methods such as random forests have shown improvements over autoregressive benchmarks [32, 31], and neural networks as well [2, 8] transformer based models are at least underexplored in the inflation literature.

In parallel, transformer-based architectures have emerged as claiming, by themselves, to have state of the art performance on several forecasting benchmarks. Studies such as Zhou et al. (2021), Wu et al. (2022), Zhou et al. (2022) argue that transformers can outperform recurrent networks and classical machine learning algorithms on benchmark datasets. At the same time, sceptical work by Zeng et al. (2023) argues that attention mechanisms may not always be well suited to temporal data.

Given the importance of inflation for monetary policy, business decisions, and household welfare, this lack of systematic evidence represents a clear gap in the literature.

This thesis addresses this gap by conducting an evaluation of transformer-based models in the context of inflation forecasting. Using a large global dataset covering 91 countries provided by Ha, Kose, and Ohnsorge (2023), Informer[48], Autoformer[46] and FEDFormer[49] will be compared with random forests and the simple linear neural networks introduced by Zeng et al. (2023). The contributions of this thesis is as follows:

1. To provide a first systematic evaluation of transformer architectures for inflation forecasting.

2. To benchmark transformer performance against state-of-the-art performing random forest and the linear benchmarks proposed by Zeng et al. (2023).

3. To contribute to the ongoing debate about the usefulness of transformers for time series forecasting by directly evaluating the claims of Zeng et al. (2023) in the context of inflation.

Thus the results will help clarify whether transformer models can be useful for inflation forecasting compared to the more lightweight machine learning benchmarks. To further investigate the claims by Zeng et al. (2023), DLinear and NLinear will also be evaluated using recursive forecasting. The recursive forecasting strategy is included in this thesis for two reasons. Zeng et al. (2023) to evaluate the claim that transformers performs better than recursive autoregressive models. And to evaluate recursive forecasting compared to the direct forecasting strategy. It is not clear whether direct or recursive forecasting is superior[44, 23] and the research investigating this recommends evaluating different forecast strategies.

The analysis will focus on key metrics root mean squared error (RMSE) and mean absolute deviation (MAD) while also considering the interpretability of transformer models.

3 Methodology

3.1 Transformer

The Transformer is an encoder–decoder neural network built from stacked attention and feed-forward layers introduced by Vaswani et al.(2017)[45]. Each encoder layer contains a multi-head self-attention sublayer followed by a position-wise feed-forward network. Each decoder layer is similar but adds a third sublayer that attends to the encoder's outputs; it also uses masking in its self-attention so that each output position can only attend to earlier positions. In effect, the encoder transforms the input sequence into continuous context vectors, and the decoder generates the output sequence one step at a time, auto-regressively consuming prior outputs when predicting the next token. The key components of this architecture are as follows. The self-attention: For each position in the input, the model computes a query, key, and value vector. The output at that position is a weighted sum of all values, where the weights come from a softmax over the query's dot-products with each key. In practice, scaled dot-product attention is used:

$$\text{Attention}(Q,K,V) = \text{softmax}\big(QK^\top/\sqrt{d_k}\big)V$$

where Q,K,V are the matrices of all queries, keys, and values, and d_k is the key dimension. This global operation means each output position can directly attend to all input positions in one step. In other words, self-attention acts like a fully-connected layer with data-dependent weights. Instead of a single attention computation, the Transformer performs several attention heads in parallel so called multi head attention. Each head applies learned linear projections to Q, K, V and computes attention independently. The outputs of all heads are then concatenated and projected back into a single vector. Vaswani et al. explain that multi-head attention "allows the model to jointly attend to information from different representation subspaces at different positions". In practice, this means different heads can focus on different types of relationships (for example, various time lags or seasonal patterns) simultaneously within each layer. Because the Transformer has no recurrence or convolution, it must inject information about the order of the sequence. This is done by adding positional encodings to the input embeddings. In the original Transformer, fixed sinusoids of varying wavelengths were used so that each position has a unique encoding (of the same dimension as the model). These encodings carry absolute and relative position information into every layer. In time series models, researchers often extend this idea: for example, learned positional embeddings or explicit timestamp features (such as hour-of-day or month-of-the-year) can be concatenated with or added to the input, encoding known calendar or trend cycles. The stacked encoder–decoder design provides flexibility. The encoder processes the entire input history in parallel through multiple attention/FFN layers to produce rich context vectors. The decoder then generates outputs step-by-step, using masked self-attention (so each step only looks at previous outputs) and encoder–decoder attention on the encoder states. This architecture is suited for forecasting: one can encode the known historical series and then have the decoder produce future values in an autoregressive (or sequence-to-sequence) manner. Notably, the decoder's masking ensures that each prediction depends only on past inputs, preserving causality.

The architecture for the transformer can be seen in Figure 1

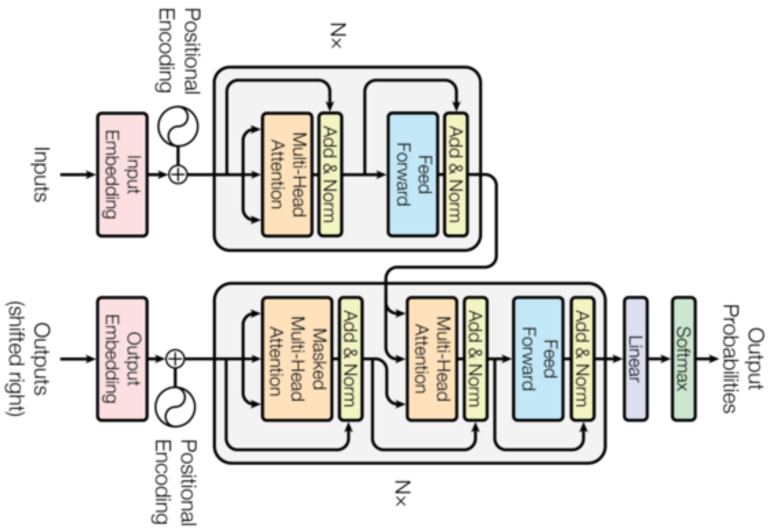


Figure 1: Transformer architecture consisting of encoder and decoder stacks. The encoder (top) applies multi-head self-attention and feed-forward layers, while the decoder (bottom) includes an additional masked multi-head attention mechanism to ensure autoregressive prediction. *Source:* Vaswani et al. (2017).

Each of these architectural features contributes to the Transformer's aims to make effective on sequential data. The global self-attention means even distant past observations can directly influence the output. Multi-head attention provides multiple views of the data (capturing diverse patterns simultaneously). Positional encodings preserve the temporal order and periodicity information. And the encoder–decoder framework supports multi-step, multivariate forecasting by design. [1] are highly recommended.

3.2 Informer

Informer, proposed by Zhou et al. (2021) [48], aim to improve the Transformer with a set of innovations tailored to long-sequence time series forecasting: ProbSparse self-attention, self-attention distillation, and a generative decoder.

Canonical self-attention computes attention scores for all query-key pairs, which scales quadratically with sequence length L, i.e., $O(L^2)$ time and space complexity. Zhou et al argues that empirical analysis of standard attention shows a long-tailed distribution of attention scores: only a few key-query pairs contribute significantly, while most pairs contribute negligibly.

[1] For more detail on the algorithms employed by Transformers the formal algorithms by Phuong and Hutter (2022)

To exploit this sparsity, Informer introduces a sparsity score for each query vector q_i, measuring how non-uniform its attention distribution is. This is formalized as:

$$M(q_i, K) = \log\left(\sum_{j=1}^{L} e^{q_i k_j^\top/\sqrt{d}}\right) - \frac{1}{L}\sum_{j=1}^{L} \frac{q_i k_j^\top}{\sqrt{d}}$$

This metric approximates the KL divergence between the query's attention distribution and a uniform distribution: higher values indicate more peaked attention, suggesting the query is more "informative."

Instead of computing attention for all queries, Informer selects only the top-u queries (where $u = c \cdot \log L$) with the highest sparsity scores to participate in full attention. This this reduces complexity. ProbSparse self-attention requires only $O(L \log L)$ time and space.

Computing exact sparsity scores still requires all dot products. To address this, Informer proposes an approximation:

$$M(q_i, K) \approx \max_j \left\{ \frac{q_i k_j^\top}{\sqrt{d}} \right\} - \frac{1}{L}\sum_{j=1}^{L} \frac{q_i k_j^\top}{\sqrt{d}}$$

This is more computationally stable (compared to log-sum-exp) and allows sampling a small subset of key-query dot-products to estimate sparsity. By focusing only on the dominant attention interactions, this approach yields attention maps that are both sparse and informative.

The Informer adopts a specialized encoder-decoder architecture designed for long-sequence time series forecasting. To support this setting, it introduces architectural refinements that allow efficient learning from long inputs while avoiding the quadratic bottleneck in self-attention.

The encoder is responsible for extracting rich representations from long input sequences. After embedding, the input time series of length L is transformed into a matrix $X^{\text{en}} \in \mathbb{R}^{L \times d_{\text{model}}}$. However, even with the ProbSparse attention mechanism, the attention outputs can remain redundant, since similar values are often aggregated repeatedly. To address this, Informer introduces a self-attention distilling procedure to reduce redundancy and memory usage while preserving important information.

Inspired by dilated convolution and pyramidal structures, Informer performs downsampling between encoder layers to progressively reduce the time dimension. After each multi-head ProbSparse attention block and subsequent feedforward processing, the output is passed through a convolutional-pooling distillation layer:

$$X^{(j+1)} = \text{MaxPool}\left(\text{ELU}\left(\text{Conv1D}\left([X^{(j)}]_{\text{AB}}\right)\right)\right)$$

Here, $[X^{(j)}]_{\text{AB}}$ includes the attention block output (attention + feedforward), Conv1D uses kernel width 3, and ELU is used as the activation function. MaxPooling (with stride 2) halves the temporal resolution at each layer.

To preserve forecasting capacity across scales, multiple distilled stacks of layers are created in parallel, each with a different number of layers (i.e., one fewer than the previous). This pyramid-like structure allows the model to maintain multi-scale temporal features. The outputs from all stacks are concatenated to form the final encoder representation.

The decoder is structured similarly to the original Transformer decoder (Vaswani et al., 2017 [45]), but it is optimized for generative inference over long sequences. It consists of stacked masked multi-head attention layers, where the masking ensures that each output timestep only attends to previous ones.

The decoder input is constructed by concatenating:

$$X^{\text{de}} = \text{Concat}(X_{\text{token}}, X_0) \in \mathbb{R}^{(L_{\text{token}}+L_y)\times d_{\text{model}}}$$

where X_{token} is a known "start token" (e.g., the previous 4 months of data), and X_0 is a placeholder (all zeros) for the future sequence to be predicted.

Masked attention is applied over this input, setting future positions in the dot-product matrix to $-\infty$, so they cannot influence earlier positions — preserving causal structure.

Informer replaces autoregressive decoding (which generates outputs one step at a time) with one-shot generative decoding. The decoder outputs the entire forecast in a single forward pass. This is made possible by conditioning on the start token and the time features of the prediction horizon embedded in X_0. For example, to predict 12 future steps (a year of monthly data), Informer uses a 4-month start token and appends the 12-length placeholder to decode all outputs simultaneously.

This significantly improves runtime efficiency and avoids the accumulation of errors common in step-wise decoding.

To summarize, the Informer architecture is shown in Figure 2

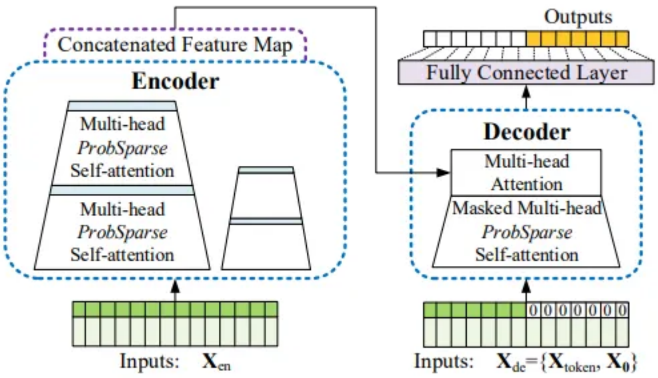
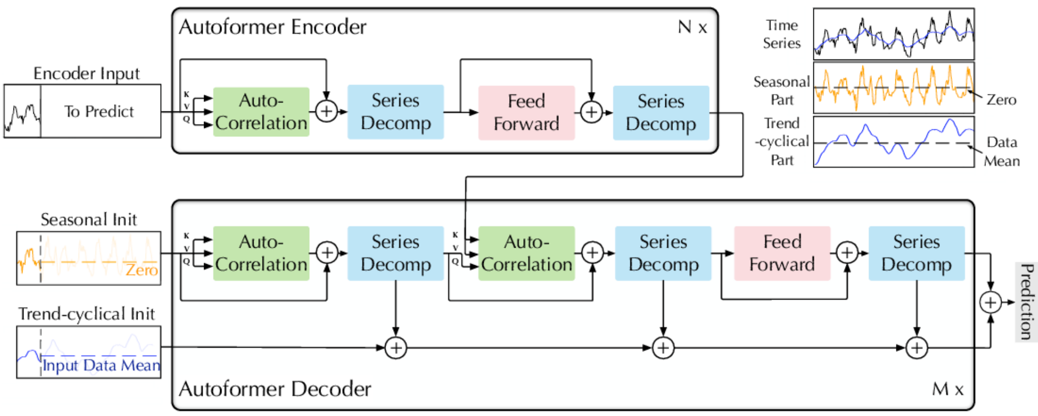


Figure 2: Architecture of the Informer model. The encoder applies stacked multi-head *Prob-Sparse* self-attention layers to compress input representations, while the decoder combines masked multi-head *ProbSparse* self-attention with standard multi-head attention to generate forecasts. A fully connected layer maps the concatenated feature representations to the output sequence. *Source:* Zhou et al. (2021).

3.3 Autoformer

Autoformer, introduced by Wu et al. (2021) [46], is a Transformer-based architecture specifically designed for long-term time series forecasting. They aim to addresses challenges in capturing intricate temporal patterns and improving computational efficiency through two primary innovations: an inner series decomposition block and an Auto-Correlation mechanism. The architecture is shown in figure 3

Figure 3: Autoformer architecture



Source: Wu et al. (2021)[46]

3.3.1 Decomposition Block

Autoformer incoorperates decomposition block as an inner operation of the model. The aim of the decomposition block is to decompose the time series, as it is typically done in time series analysis, in to three parts: trend-cycle, seasonal variation, and random fluctuations. This internal decomposition enables the model to extract long-term stationary trends from intermediate hidden variables progressively. Specifically, Autoformer employs a moving average to smooth out periodic fluctuations, thereby highlighting long-term trends. For an input series $X \in \mathbb{R}^{L\times d}$ where L is length of the series, the process is defined as:

$$X_t = \text{AvgPool}(\text{Padding}(X)) \tag{1}$$

$$X_s = X - X_t \tag{2}$$

Here, X_t represents the extracted trend component, and X_s denotes the seasonal component. Where AvgPool is a moving average with a given kernel size, Padding is simply just a 1-padding on each side of the series entirely defined by the kernel size.

3.3.2 Auto Correlation (Attention)

Autoformer captures period-based dependencies in time series forecasting by using a Auto-Correlation mechanism, which expands information utilization through time-delay similarities. Unlike self-attention, which models point-wise relationships, Auto-Correlation detects periodic dependencies by computing autocorrelation on the input series. Autoformer achieves this by replacing the standard dot-product similarity in self-attention with autocorrelation between the query Q and key K. Query and key are projected representations of the time series, while value V represents the sequence data being aggregated across time.

For a discrete-time process $\{X_t\}$, the autocorrelation function is defined as:

$$R_{XX}(\tau) = \lim_{L\to\infty} \frac{1}{L} \sum_{t=1}^{L} X_t X_{t-\tau}$$

where $R_{XX}(\tau)$ quantifies the similarity between $\{X_t\}$ and its lagged counterpart $\{X_{t-\tau}\}$.

To efficiently compute autocorrelation, Autoformer applies the Fast Fourier Transform (FFT), which accelerates the identification of dominant periodicities. These estimated period lengths, denoted as $\tau_1,\ldots,\tau_k$, are selected using:

$$\tau_1,\ldots,\tau_k = \text{argTop}_k\left(R_{Q,K}(\tau)\right)$$

Where Top_k selects the k most prominent delays based on their autocorrelation scores and Q denotes the query and K the key.

Once key periodic time delays are identified, the Time Delay Aggregation (TDA) module aligns similar sub-series at the same phase positions by rolling the series based on the selected delays. The aggregation process is defined as:

$$\hat{R}_{Q,K}(\tau_1),\ldots,\hat{R}_{Q,K}(\tau_k) = \text{SoftMax}\left(R_{Q,K}(\tau_1),\ldots,R_{Q,K}(\tau_k)\right) \tag{3}$$

$$\text{Auto-Correlation}(Q,K,V) = \sum_{i=1}^{k} \text{Roll}(V,\tau_i)\hat{R}_{Q,K}(\tau_i) \tag{4}$$

where $\text{Roll}(V,\tau_i)$ shifts the series by τ_i, reintroducing the shifted-out elements at the opposite end. Softmax normalization assigns appropriate weights to each time delay.

For a multi-head setup with h attention heads and hidden dimension d_{model}, input series representations are divided into multiple subspaces:

$$Q_i, K_i, V_i \in \mathbb{R}^{L\times \frac{d_{\text{model}}}{h}}, \quad i \in \{1,\ldots,h\}$$

Each head applies the Auto-Correlation operation independently:

$$\text{MultiHead}(Q,K,V) = W_{\text{output}} \cdot \text{Concat}(\text{head}_1,\ldots,\text{head}_h)$$

where:

$$\text{head}_i = \text{Auto-Correlation}(Q_i, K_i, V_i)$$

This structure enables Autoformer to capture dependencies across multiple time scales efficiently.

The final forecast is obtained by summing the seasonal and trend components, both refined through the Auto-Correlation mechanism:

$$\hat{Y} = W_S \cdot X_{\text{de}}^M + T_{\text{de}}^M$$

where W_S projects the deep transformed seasonal component X_{de}^M into the target dimension, and T_{de}^M represents the trend component.

By focusing on series-wise dependencies rather than point-wise attention, Autoformer aims to capture long-range periodic structures, improving forecasting performance over conventional self-attention-based models.

3.4 FEDformer

Like Autoformer, FEDformer is designed to handle long-term dependencies in time series forecasting, and by leveraging ideas from classical time series decomposition and spectral analysis.

FEDformer introduced by Zhou et al. 2022 [49] shifts more fundamentally into the frequency domain. It applies the Discrete Fourier Transform (DFT) directly to the residual (seasonal) component and performs frequency selection and enhancement, retaining only dominant periodicities before reconstructing the signal via the inverse DFT. This approach enables FEDformer to model global periodic structures more efficiently and reduces the reliance on full attention mechanisms.

FEDformer's architecture centers around three key components: a Frequency Enhanced Decomposition block that processes the seasonal part of the series in the frequency domain, an efficient frequency-based attention mechanism to reduce computational complexity, and separate modeling of trend and seasonal components to capture different aspects of the time series.

The Frequency Enhanced Decomposition (FED) block leverages the Discrete Fourier Transform (DFT) to extract and enhance dominant frequency components from the time series seasonal input. Let $x_n \in \mathbb{R}^D$ for $n = 0, 1, \ldots, N-1$ denote the input sequence in the time domain, where N is the sequence length and D the feature dimension.

The DFT transforms x into the frequency domain as:

$$X_l = \sum_{n=0}^{N-1} x_n e^{-i\omega l n}, \quad l = 0, 1, \ldots, L-1,$$

where i is the imaginary unit, $\omega = \frac{2\pi}{N}$, and $L \leq N$ is the number of frequency bins retained. The inverse DFT reconstructs the time-domain signal from the frequency representation:

$$x_n = \sum_{l=0}^{L-1} X_l e^{i\omega l n}, \quad n = 0, 1, \ldots, N-1.$$

Direct computation of the DFT has complexity $O(N^2)$, which is reduced to $O(N \log N)$ via the Fast Fourier Transform (FFT). FEDformer further reduces complexity to approximately $O(N)$ by selecting a subset of $M \ll N$ dominant frequency modes.

The process within the Frequency Enhanced Block (FEB-f) is as follows:

The input $x \in \mathbb{R}^{N\times D}$ is first projected using a learnable weight matrix $W \in \mathbb{R}^{D\times D}$, producing $q = xW$. Then the projected signal q is transformed to the frequency domain:

$$Q = \mathscr{F}(q) \in \mathbb{C}^{N\times D}.$$

Then a selection operator retains only the M most informative modes:

$$\widetilde{Q} = \text{Select}(Q) \in \mathbb{C}^{M \times D}, \quad M \ll N,$$

Then the retained frequencies are modulated by a learnable complex-valued kernel $R \in \mathbb{C}^{D \times D \times M}$, performing a weighted combination of channels:

$$Y_{m,d_o} = \sum_{d_i=1}^{D} \widetilde{Q}_{m,d_i} \cdot R_{d_i,d_o,m},$$

where d_i and d_o index input and output feature channels, respectively. After this the enhanced frequencies Y are zero-padded back to size N and transformed to the time domain via inverse FFT:

$$\text{FEB-f}(q) = \mathscr{F}^{-1}(\text{Padding}(Y)) \in \mathbb{R}^{N \times D}.$$

By focusing attention on selected dominant frequencies and parameterizing their enhancement, the FEB-f Zhou et al. argue that this block effectively captures and reconstructs the important seasonal patterns while keeping computational demands manageable. Pseducode for the algorithm can be found in the appendix 1

Moreover, Zhuo et al. introduces Frequency Enhanced Attention with Wavelet Transform (FEA-w) module that replaces the canonical dot-product attention with a wavelet-based alternative, allowing the model to capture localized time-frequency interactions. This is particularly beneficial in multidimensional settings, where different variables may exhibit localized patterns at different temporal scales.

FEA-w modifies the standard transformer attenting by using a shared decomposition matrix to break q, k, and v into multiple frequency components at multiple resolutions. At each decomposition level L, three signals are extracted. low-frequency (trend-like) component: $s\_n^{(L)}$, high-frequency (detail) component: $d\_n^{(L)}$, residual for the next decomposition level: $x^{(L+1)}$

Each of these components is then processed by shared frequency enhanced attention blocks (FEA-f), applied in parallel. The FEA-f block operates similarly to its Fourier-based counterpart, but over the wavelet-decomposed signals.

Let $q^{(L)}, k^{(L)}, v^{(L)}$ denote the signals at decomposition level L. The FEA-w processing proceeds in three main stages.

First, each of $q^{(L)}, k^{(L)}, v^{(L)}$ is decomposed using a fixed wavelet basis—such as Legendre multiwavelets—into a low-frequency part $s_q^{(L)}, s_k^{(L)}, s_v^{(L)}$, a high-frequency part $d_q^{(L)}, d_k^{(L)}, d_v^{(L)}$, and a residual that becomes the input at the next level, $q^{(L+1)}, k^{(L+1)}, v^{(L+1)}$.

Next, each component is passed through an FEA-f module to apply attention in the wavelet domain. For example, the attention output on the high-frequency part is computed as

$$Y_d^{(L)} = \mathscr{F}^{-1}\left(\text{Padding}\left(\sigma\left(\widetilde{Q}_d \cdot \widetilde{K}_d^{\top}\right) \cdot \widetilde{V}_d\right)\right),$$

where $\widetilde{Q}_d = \text{Select}(\mathscr{F}(d_q^{(L)}))$, and similarly for $\widetilde{K}_d$ and $\widetilde{V}_d$. Here σ is an element-wise activation function, such as Softmax or tanh, and the selection step retains the M most dominant wavelet frequency modes.

After the final decomposition level $L_{\max}$, the outputs from each level—namely $Y_s^{(L)}$, $Y_d^{(L)}$, and $Y_r^{(L+1)}$—are recursively combined to reconstruct the signal. This reconstruction process doubles the length at each step, ultimately recovering the full-length signal in the time domain.

The complete FEA-w block is thus defined recursively as

$$\text{FEA-w}(q,k,v) = \text{Reconstruct}\left(\left\{\text{FEA-f}(d^{(L)}), \text{FEA-f}(s^{(L)}), \text{FEA-f}(q^{(L+1)})\right\}_{L=1}^{L_{\max}}\right).$$

Similarly FEDformer employs a Frequency Enhanced Decomposition block using wavelet transform (FEA-w).

Zhou et al. recommends the wavelet transform for multivariate problems.

While FEA-w efficiently captures multiscale dependencies via localized, frequency-aware attention, modeling long-range dependencies and diverse temporal behaviors across variables often requires additional flexibility. This is adressed by using a Mixture-of-Experts (MoE) layer on top of the FEA-w outputs.

The MoE mechanism enables dynamic routing of input signals to specialized expert networks. Let $z \in \mathbb{R}^{T \times d}$ denote the sequence produced by the final FEA-w reconstruction, where T is the sequence length and d the feature dimension. A gating network computes soft routing weights $G \in \mathbb{R}^{T \times E}$, where E is the number of experts. These weights determine the contribution of each expert to the final output at every timestep:

$$G = \text{Softmax}(W_g z + b_g),$$

where W_g and b_g are learnable parameters of the gating function.

Each expert $\mathscr{E}_i$, for $i = 1, \ldots, E$, is a lightweight feed-forward network (e.g., a two-layer MLP with nonlinear activation), and the output is given by:

$$y_t = \sum_{i=1}^{E} G_{t,i} \cdot \mathscr{E}_i(z_t),$$

for each timestep $t \in \{1, \ldots, T\}$. This architecture allows the model to allocate specialized processing capacity dynamically, depending on the local structure and frequency content of the input sequence.

In practice, sparsity constraints (e.g., top-k gating) can be applied to the routing weights G to reduce computational overhead and encourage specialization.

The integration of MoE with wavelet-enhanced attention allows the model to combine coarse-to-fine signal decomposition with conditional computation, resulting in a powerful and flexible architecture for modeling complex temporal patterns in multivariate time series.

To summarize the Architecture of FEDFormer is shown in Figure 4

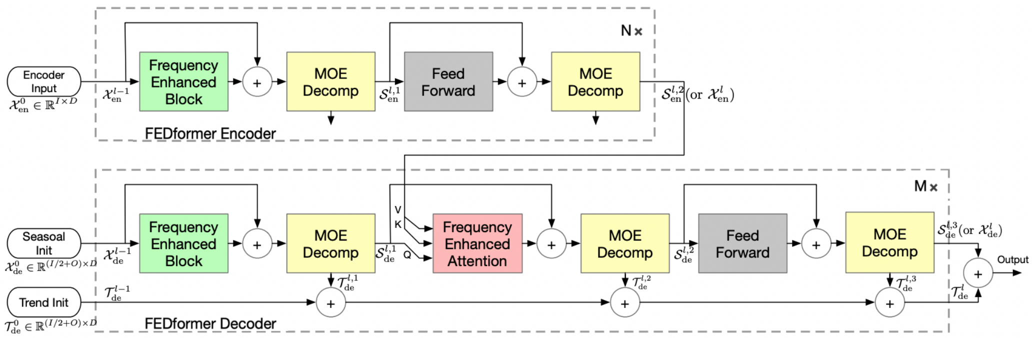


Figure 4: FEDFormer architecture. The encoder applies frequency-enhanced blocks and decomposition layers (MOE Decomp), followed by feed-forward networks. The decoder combines seasonal and trend initializations, applying frequency-enhanced attention together with decomposition to produce the output sequence. *Source:* Zhou et al. (2022).

3.5 Dlinear and Nlinear

Nlinear and Dlinear was first introduced by Zeng et al. (2023) [47], as, in their own words, embarrassingly simple baselines for time series forecasting. They argue that traditional Transformer-based forecasting models often compare against iterative multistep baselines, which suffer from error accumulation. Instead, these baselines follows a direct multi-step approach, which forecasts all future time steps in one step. This has several advantages, it is "plug and play" baselines compared to the transformer models, no need to do another forecasting strategy and no need to handle multiple outputs any different

DLinear utilises the decomposition scheme used by Autoformer. It first decomposes X as shown in Equation 1. X_t and X_s are then independently processed through a linear one-layer layer, and the outputs are summed to form the final prediction. The aim is to improve forecast accuracy by explicitly modeling the trend for datasets with clear long-term patterns compared to a vanilla linear neural network with no decomposition.

NLinear was build to better handle distribution shifts by applying a simple normalization step. The method begins by subtracting the last value of the input sequence from all elements, after which the normalized sequence is passed through a linear layer. Finally, the last value is added back to form the prediction. This procedure centers the sequence around its most recent observation, which helps the model cope with changing distributions while still preserving the original scale of the output.

3.5.1 Random Forest

Random Forest (RF) is used as a baseline model, as it has shown state of the art performance in inflation forecasting [32, 31]. RF , popularized by Breiman (2001) [11], are an ensemble method that combines many decision trees to reduce prediction variance and improve generalization. They build on (bootstrap aggregation), where each tree is trained on a bootstrap sample of the data. For regression, predictions are averaged across trees:

$$\hat{f}_{\mathrm{rf}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x),$$

where $T_b(x)$ is the prediction of the b-th tree. For classification, the prediction is based on majority vote.

Random forests improve on bagging by decorrelating trees: at each split, a random subset of features is considered. This reduces the pairwise correlation between trees, which limits the variance reduction in bagging.

Besides inflation forecasting, random forests are widely used due to their accuracy, robustness, and minimal tuning.

3.6 Model setup

3.6.1 Single target model

The forecasting problem can be written as

$$\pi^h_{i,t+h} = G_{i,t,h}\left(\pi^{AR}_{i,t-p+1:t}, \gamma_{i,t}, g_t, x_{i,t}\right) + u_{i,t+h},$$

where i indexes countries and h the forecast horizon. The dependent variable $\pi^h_{i,t+h}$ is the standard inflation rate for horizon h, defined as $\pi^h_{i,t+h} = \log(P_{t+h}/P_{t+h-1})$ where P_t denotes the headline CPI level. Predictors consist of the last p lags of domestic inflation, a set of monthly dummies capturing seasonal effects, a global inflation factor, and the contemporaneous inflation series of all other countries in the panel. The error term $u_{i,t+h}$ is assumed to have zero mean.

Future inflation is thus modeled as a potentially nonlinear function of both domestic and international determinants. Preliminary checks indicated that including lags of the global factor or of foreign inflation series did not improve forecast accuracy, so only contemporaneous values are considered. In total, the forecasting design allows for 108 predictors.

The direct forecast equation is given by:

$$\hat{\pi}^{h}_{i,t+h} = \hat{G}_{i,t-R+1:t,h}\left(\pi^{AR}_{i,t-p+1:t}, \gamma_{i,t}, g_t, x_{i,t}\right),$$

where the mapping function $\hat{G}_{i,t-R+1:t,h}$ is estimated using a rolling window of length $R = 360 - p + 1$. For the single target models all forecasts are produced separately for each country–horizon pair.

The inclusion of a global factor follows Medeiros et al. (2023)[31], who show that most of the predictive information in a high-dimensional panel of inflation series can be captured by global and regional factors. In the present study, regional factors are not included because, given the structure of the multidimensional models, incorporating them would either require using all regional factors simultaneously or none, while single-target models would only use the regional factor corresponding to the country being forecast. To maintain interpretability, the analysis therefore focuses exclusively on the global factor.

An additional motivation for specifying the forecasting problem at the monthly horizon is that this formulation directly aligns with the frequency of the data and allows flexible evaluation of both short- and long-horizon forecasts. Month-by-month modeling is often used in empirical macro forecasting. [29, 5]. Although policymakers usually communicate inflation projections in terms of year-over-year rates or annual averages, the underlying forecasting exercises at central banks and in academic work are often performed at the monthly or quarterly frequency [43, 7]. Framing the problem at the monthly level therefore facilitates comparability with the broader forecasting literature, while aggregated cumulative measures can subsequently be constructed. However if the cumulative inflation is the target of interest a better approach is of course to model these directly.

3.6.2 Multi Target Model

In the multi-target setup, inflation is forecast simultaneously for all countries. Let $\boldsymbol{\pi}^{h}_{t+h} = (\pi^{h}_{1,t+h}, \dots, \pi^{h}_{n,t+h})'$ denote the vector of inflation rates at horizon h. The forecasting problem is written as

$$\boldsymbol{\pi}^h_{t+h} = \mathbf{G}_{t,h}\Big(\boldsymbol{\pi}^{AR}_{t-p+1:t}, \boldsymbol{\gamma}^*_t, \mathbf{g}_{t-p+1:t}\Big) + \mathbf{u}_{t+h},$$

where $\boldsymbol{\pi}^{AR}_{t-p+1:t}$ collects the p most recent lags of inflation for every country in the panel, $\mathbf{g}_{t-p+1:t}$ contains the p lagged values of the global inflation factor, and $\boldsymbol{\gamma}^*_t$ denotes the monthly dummy variables (starred to indicate they are treated differently across models). The error vector $\mathbf{u}_{t+h}$ is assumed to have zero mean.

Including lags of all countries' series and the lagged global factor allows the model to exploit temporal dependencies both within and across countries. The direct multi-target forecast is therefore

$$\hat{\boldsymbol{\pi}}^h_{t+h} = \hat{\mathbf{G}}_{t-R+1:t,h}\Big(\boldsymbol{\pi}^{AR}_{t-p+1:t}, \boldsymbol{\gamma}^*_t, \mathbf{g}_{t-p+1:t}\Big),$$

with $\hat{\mathbf{G}}_{t-R+1:t,h}$ estimated on a rolling window of length R.

Transformer-based models and use positional encodings to inject sequence order information [45]. Time/frequency information is furthermore handled in a model-specific manner. Informer augments the input embedding with explicit time-stamp (calendar) embeddings (local and global features such as minute, hour, week, month, holiday), providing calendar signals directly in the input representation [48], Autoformer incorporates a progressive series-decomposition block together with an auto-correlation mechanism to extract trend and seasonal components within the network [46]; FEDformer combines decomposition with frequency-domain operations (e.g., Fourier / sparse frequency projection) to capture global periodic structure [49]. Because these Transformer variants therefore implement frequency-awareness either via input time-stamp embeddings or internal decomposition/frequency modules, the monthly dummy variables $\boldsymbol{\gamma}^*_t$ are omitted for Informer, Autoformer and FEDformer in this study; $\boldsymbol{\gamma}^*_t$ is supplied explicitly only to architectures that lack such built-in mechanisms i.e. DLinear.

In the recursive setup[2], the models are trained for one-step-ahead prediction and multi-step forecasts are generated by iteratively feeding back past predictions as inputs. Formally, let $\boldsymbol{\pi}_{t+1} = (\pi_{1,t+1}, \ldots, \pi_{n,t+1})'$ denote the vector of inflation rates one period ahead. The forecasting problem is

$$\boldsymbol{\pi}_{t+1} = \mathbf{G}^{rec}_t\Big(\boldsymbol{\pi}^{AR}_{t-p+1:t}, \boldsymbol{\gamma}^*_t, \mathbf{g}_{t-p+1:t}\Big) + \mathbf{e}_{t+1},$$

where the notation follows the multi-target case: $\boldsymbol{\pi}^{AR}_{t-p+1:t}$ collects the p most recent lags of all countries' inflation, $\mathbf{g}_{t-p+1:t}$ contains the lagged global factor, and $\boldsymbol{\gamma}^*_t$ denotes the monthly dummies.

[2]Note that this is also often called iterative forecasting

A forecast h-steps ahead is then obtained recursively as

$$\hat{\boldsymbol{\pi}}_{t+h} = \hat{\mathbf{G}}^{rec}_{t-R+1:t}\left(\hat{\boldsymbol{\pi}}^{AR}_{t-p+1:t+h-1}, \boldsymbol{\Upsilon}^{*}_{t+h-1}, \hat{\mathbf{g}}_{t-p+1:t+h-1}\right),$$

where lagged values for horizons beyond t are replaced with previously generated forecasts.

In this thesis, the recursive approach is implemented with DLinear and NLinear. Since only the inflation rate of each country is being forecasted, the global factor is simply the mean of the forecasted inflation rates.

The Estimation for all of the above models is performed on a rolling window of length R, and out-of-sample forecasts cover March 2010–December 2019.

3.7 Data

The analysis relies on the Global Database on Inflation compiled by Ha, Kose, and Ohnsorge (2023), and follows the data preparation procedure in Medeiros et al. (2023)[31]. The database reports inflation at monthly, quarterly, and annual frequencies for a wide set of countries. Monthly headline inflation is used, as it maximizes the number of observations and provides the broadest coverage of goods and services.

The headline inflation is defined as:

$$\pi_t = \log(P_t) - \log(P_{t-1}), \tag{5}$$

where π_t denotes the monthly headline inflation rate in period t, and P_t is the headline the Consumer Price Index (CPI), headline inflation [25].

To construct a balanced panel, the sample is restricted to the period January 1980–December 2019, yielding 480 monthly observations. Countries with more than two consecutive missing values or more than four missing values in total are excluded. Remaining gaps are interpolated using the expectation–maximization algorithm proposed by Stock and Watson (2002a)[41], resulting in only four imputed values across the entire dataset (two for Suriname, one for Myanmar and one for Saudi Arabia). The final panel covers 91 countries.

The dataset exhibits two notable features. First, several countries experienced episodes of very high inflation during the early part of the sample. To account for this, the monthly inflation series of Bolivia, Brazil, and Peru. Hyperinflation is defined, following Cagan (1956) [12], as a monthly inflation rate of at least 50 percent in one or more months during the sample period.

Second, inflation dynamics vary considerably in terms of seasonality. Some countries display strong seasonal cycles, while others show little or no seasonal pattern. Rather than applying country-specific seasonal adjustments, the analysis uses raw (non-seasonally adjusted) inflation data and includes monthly dummy variables as additional predictors. This allows the models to capture and evaluate each country's unique seasonal structure within the forecasting exercise.

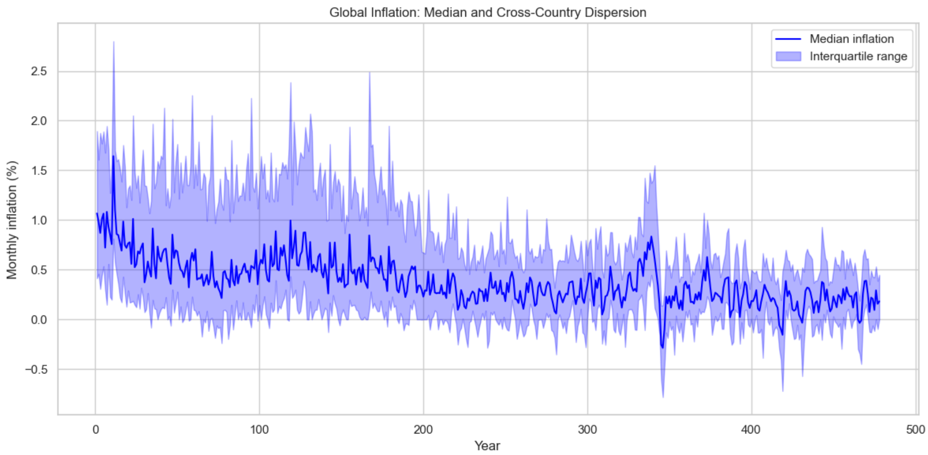


Figure 5: Median monthly inflation across 91 countries (blue line) with interquartile range (shaded area), 1980–2019.

Figure 5 illustrates the cross-country dynamics of inflation over the sample period. The median monthly inflation has trended downward since the early 1980s, stabilizing at relatively low levels from the 2000s onwards. At the same time, the interquartile range narrowed considerably, indicating a reduction in cross-country dispersion. The early part of the sample is characterized by pronounced volatility and high cross-country heterogeneity, while the period after the Global Financial Crisis shows both lower inflation and reduced dispersion.

3.8 Forecast Evaluation Methods

Forecast accuracy is assessed using the root mean squared error (RMSE) and the median absolute deviation (MAD). The RMSE is the square root of the average squared forecast error, expressed in the same units as the forecasted variable. It gives greater weight to large errors and is therefore sensitive to outliers. The MAD is the median of the absolute deviations of forecast errors from their median, offering a robust measure of typical forecast accuracy that is less affected by extreme values.

For country c and horizon h, the two measures are defined as

$$\mathrm{RMSE}_{h,c} = \sqrt{\frac{1}{T_h}\sum_{t=1}^{T_h} \hat{e}_{t,h,c}^2}, \tag{6}$$

$$\mathrm{MAD}_{h,c} = \mathrm{median}\left(\left|\hat{e}_{t,h,c} - \mathrm{median}(\hat{e}_{t,h,c})\right|\right), \tag{7}$$

where $\hat{e}_{t,h,m} = \pi_t^h - \hat{\pi}_{t,m}^h$ denotes the forecast error, π_t^h is the realized value, $\hat{\pi}_{t,m}^h$ the forecast, and T_h the number of evaluation observations at horizon h.

The average superior predictive ability (aSPA) test from Quaedvlieg (2021) is used to conduct multi-horizon forecast comparisons. For any two models i and j, and horizon h, define the loss differential as

$$d_{ij,t}^h = L\left(y_{t+h}, \hat{y}_{i,t}^h\right) - L\left(y_{t+h}, \hat{y}_{j,t}^h\right). \tag{8}$$

Next compute the weighted average loss differential:

$$\mu_{ij}^{(\mathrm{Avg})} = \sum_{h=1}^{H} w_h \frac{1}{T}\sum_{t=1}^{T} d_{ij,t}^h, \tag{9}$$

where the weights $w_1, \ldots, w_H$ sum to one. In this thesis equal importance is placed on each horizon by setting $w_h = 1/H$.

The null hypothesis of the aSPA test is that the weighted average loss differential is less than or equal to zero, with the alternative being that it is greater than zero:

$$H_{0,\mathrm{aSPA}} : \mu_{ij}^{(\mathrm{Avg})} \leq 0, \quad H_{A,\mathrm{aSPA}} : \mu_{ij}^{(\mathrm{Avg})} > 0. \tag{10}$$

Testing this hypothesis is done using a Student's t-statistic of the form

$$t_{ij}^{\mathrm{aSPA}} = \frac{\sqrt{T}\,\mu_{ij}^{(\mathrm{Avg})}}{\sqrt{\hat{\omega}_{ij}}}, \tag{11}$$

where $\hat{\omega}_{ij}$ is estimated using any consistent Heteroskedasticity and Autocorrelation Consistent estimator, here the quadratic spectral kernel [3] is used.

As special weight choices may require nonstandard critical values, and bootstrap methods often yield better finite-sample performance, Quaedvlieg (2021) is followed by obtaining critical values using a moving block bootstrap with block length three and 999 resamples.

The aSPA test is essentially a Diebold and Mariano (1995) test applied to the weighted average loss differential across horizons. Also note that for a single horizon the aSPA reduces to the standard Diebold and Mariano test

Complex machine learning models can achieve high predictive accuracy, but their internal mechanisms are often difficult to interpret. In economic forecasting, interpretability is essential to assess which factors drive the predictions. To provide such insights, SHapley Additive exPlanations (SHAP; Lundberg and Lee (2017)) are employed to decompose model forecasts into contributions from individual features and to derive measures of variable importance. SHAP is often used together with machine learning models in macro economic forecasting (including inflation). [31, 4, 13]

SHAP is based on the concept of Shapley values (shap values) from cooperative game theory [38]. In a cooperative game with a set of players $N = \{1, 2, \ldots, M\}$, the Shapley value of player $i \in N$ with respect to a value function $v : 2^N \to \mathbb{R}$ is defined as

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{12}$$

This expression assigns to each player the average marginal contribution it makes when joining all possible subsets of players, with weights ensuring a fair allocation across all possible orderings.

In the machine learning context, the "players" are the input features, and the "value function" corresponds to the expected prediction conditional on a subset of features. For a model $f(x)$ with input vector $x \in \mathbb{R}^M$, the Shapley value ϕ_i represents the contribution of feature i to the prediction $f(x)$. Predictions can thus be decomposed as

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i, \tag{13}$$

where $\phi_0 = \mathbb{E}[f(x)]$ is the expected prediction across the dataset. This additive decomposition satisfies efficiency (the contributions sum to the prediction), symmetry (features with identical contributions receive equal values), and consistency (if a feature's marginal contribution increases in a new model, its Shapley value does not decrease).

Besided explaining individual forecasts, SHAP values can be aggregated across the sample to obtain variable importance measures. Here the absolute SHAP value for each feature is computed:

$$I_i = \frac{1}{T} \sum_{t=1}^{T} |\phi_i^{(t)}|, \tag{14}$$

where $\phi_i^{(t)}$ is the SHAP value of feature i for forecast t, and T is the number of forecasts. In practice, the aggregation can be defined over any relevant dimension, such as time, forecast horizon, country, continent, etc. depending on the desired level of analysis.[3]

3.9 Tree parzen estimator for hyperparameter searches

Bayesian optimization is a sequential strategy for tuning hyperparameters of black-box functions by constructing a probabilistic surrogate model from past evaluations and selecting new candidates via an acquisition function. A common choice is Expected Improvement (EI), defined for minimization as

$$EI(x) = \mathbb{E}\left[\max(0, y^* - f(x))\right], \tag{15}$$

where y^* is the best observed value [33]. Traditional approaches employ Gaussian Processes (GPs) to model $p(y \mid x)$, but these scale as $\mathcal{O}(n^3)$ with the number of evaluations and are poorly suited to categorical or conditional parameters [39].

The Tree-structured Parzen Estimator (TPE) [10] instead models the inverse distribution $p(x \mid y)$. A quantile threshold y^* is chosen such that $P(y < y^*) = \gamma$, splitting observations into a "good" set $\{x_i : y_i < y^*\}$ and a "bad" set $\{x_i : y_i \geq y^*\}$. Two density functions are constructed:

$$l(x) = P(x \mid y < y^*), \tag{16}$$

$$g(x) = P(x \mid y \geq y^*). \tag{17}$$

Using Bayes' rule, the expected improvement can be expressed up to a constant as

$$EI(x) \propto \frac{l(x)}{g(x)}, \tag{18}$$

so TPE samples candidates from $l(x)$ and selects those maximizing $l(x)/g(x)$, thereby favoring regions associated with past good performance while avoiding those typical of poor outcomes.

Because densities are constructed conditionally, TPE naturally supports tree-structured hyperparameter spaces. Advantages include computational scalability (linear-time density updates versus cubic GP fitting), flexibility with heterogeneous hyperparameters, and adaptive exploration–exploitation balance controlled by γ. Limitations are weaker handling of strong interactions between hyperparameters due to independence assumptions and sensitivity to the choice of γ. TPE has been widely adopted in practice, most

[3]SHAP values has been estimated in Python using the shap package

notably as the default hyperparameter optimization algorithm in the Optuna framework [1], which provides an efficient implementation capable of scaling to large search spaces.

4 Results and discussion

4.1 Parameters tuned

The choice of hyperparameter search spaces is partly inspired by Zhou et al. (2021) [48], who tune for sequence length, model depth, and attention heads. While following the same principle of balancing model expressiveness against efficiency, this thesis adapts the ranges to the monthly inflation forecasting setting. In particular, the sequence and label lengths are restricted to 12–36 months, corresponding to one to three years of lagged information, rather than the much longer horizons used in high-frequency datasets frequently used in the transformer forecasting literature [48, 46, 49, 47]. The ranges for embedding dimension, feed-forward size, and dropout are similarly adjusted to reflect the scale of the data. Note that while Zhou et al. (2021) mentions a grid ,there is none to be seen in the GitHub linked to the paper. Furthermore neither Wu et al. (2022) and Zhou et al. (2022) mentions anything about a hyperparameter search, this is implemented here since during initial test the transformer models show terrible performance and severe overfitting problem if the model was not controlled in any way.

The common hyperparameters in Table 1 are tuned across all transformer models. The number of attention heads and encoder layers control the representational capacity: more heads enable the capture of diverse temporal dependencies, while deeper encoders increase abstraction at the cost of computation. Embedding and feed-forward dimensions determine the richness of the learned representations, with ranges chosen to span lightweight to high-capacity variants without exceeding practical limits. Sequence and label lengths govern how much historical information is used as input and how much context is retained for decoding, with only valid pairs considered where the label length does not exceed the sequence length. Regularization is controlled through dropout in [0.0, 0.3], as stronger dropout often harms time series performance. Optimization hyperparameters follow standard practice: a log-uniform learning rate in [1e-5, 1e-3] and batch sizes 8, 16, 32 to balance gradient stability against memory constraints.

In addition, Autoformer and FEDformer include model-specific hyperparameters (Table 1). The kernel size sets the moving-average window in the seasonal–trend decomposition, where smaller windows capture short-term seasonalities and larger windows allow longer cycles. FEDformer further includes a choice between Fourier and wavelet representations. Both are considered, as Fourier transforms are efficient for global periodic patterns, while wavelets can adapt to local and nonstationary structures.

The rest of the hyperparameters of the models are kept at the default setting used for the respective model. Given this large search space the hyperparameters for the transformer models were tuned using Bayesian optimization via the Tree-structured Parzen Estimator (TPE) algorithm.[4]

Hyperparameter	**Search space**	**Notes**
Common hyperparameters		
`n_heads`	{8,16}	Number of attention heads.
`e_layers`	{2,3,4}	Number of encoder layers.
(seq_len, label_len)	{12,24,36}, label_len ≤ seq_len	Only valid sequence–label pairs considered.
`d_model`	{128,256,512}	Embedding dimension.
`d_ff`	{256,512,1024,2048}	Feed-forward network dimension.
`dropout`	[0.0,0.3]	Dropout rate.
`learning_rate`	Log-uniform [1e-5,1e-3]	Optimized on log scale.
`batch_size`	{8,16,32}	Mini-batch size.
Model-specific hyperparameters (Autoformer / FEDformer)		
`kernel_size`	{5,9,13,25}	Window size of moving average for trend/seasonal decomposition.
`version`	{Fourier, Wavelets}	Spectral transform choice (FEDformer).

Table 1: Hyperparameters tuned for transformer models.

For the DLinear model, hyperparameters were tuned using a grid search over kernel size $5, 9, 13$ and input chunk length (lags) $4, 8, 12, 24$. NLinear is only tuned one the input chunk length which is the same as for DLinear.

The Random Forest benchmark has also been tuned using a grid search. The number of autoregressive lags p was selected from $4, 12, 24$, and the number of predictors considered at each split was tuned using values motivated by Medeiros, Schütte, and Soussi (2023) is $[p, 2/3p, 1/3p, \sqrt{p}, \log_2 p]$.

For all hyperparameter searches a walk-forward cross-validation scheme is used with 5 folds. For each rolling window this produces five expanding-window folds: in fold k the training set contains all observations up to the start of the k-th test block and the test set is the immediately following. Hyperparameter selection is carried out within each rolling window using these five folds; the configuration with the best average cross-validated mean squared error is refit on the full window and used for forecasting until the next re-tuning. Tuning is performed in the first training window and subsequently every December for every model besides RF. Since the RF procedure consists of 91 individual models it was not feasible for this thesis to

[4]The optuna package[1] has been used for TPE

tune the model every December, instead each model is tuned twice once in the start of the validation period and once halfway through.

For recursive models, forecasts are generated step by step: the model produces a one-step-ahead prediction, which is then appended to the input series and used to generate the next forecast. During cross-validation, this recursive procedure is applied within each fold so that multi-step predictions are based only on information available up to that point. Meaning that for each target in the validation fold, the model conditions on the true historical data up to the start of the fold and thereafter on its own past predictions until the horizon is reached. This practice is recommended in the forecasting literature, since the validation procedure should mirror the way forecasts are produced in practice [9]. [5]

4.2 Empirical Forecast Performance

Tables 10 report the main forecasting results, summarizing performance in terms of Root Mean Squared Error (RMSE) and Median Absolute Deviation (MAD) across horizons $h = 1, \ldots, 12$. This is averages across countries are emphasized rather than country-level results because it would be an information overload if the performance for each country would be reported. The tables highlight the minima at each horizon with a box, the second smallest in bold and third smallest with an underscore.

From the table it is apparent that the recursive linear neural networks (NLinear recursive and DLinear recursive) perform surprisingly well. In terms of RMSE, NLinear recursive achieves the lowest error across nearly all horizons from $h = 2$ to $h = 11$, with DLinear often ranking second. The same pattern holds for MAD, where NLinear recursive delivers the most accurate forecasts from $h = 5$ to $h = 11$. This result is similar to Marcellino, Stock, and Watson (2006), who show that iterated forecasts often outperform direct forecasts in U.S. macroeconomic data. And also shows yet another purely linear model that is hard to beat. Random Forests (RF) do however still perform strongly. For both RMSE and MAD, RF achieves the lowest or second-lowest error at $h = 1$ and again at the longest horizon ($h = 12$). This still shows the strength of random forest as show by Medeiros et al. (2021) and Medeiros, Schütte, and Soussi (2023), but with the good performance of the linear neural networks it questions the claims about the importance of non-linear models

[5] Implementation detail: For the transformer models the models were build using the GitHub directory by Zhou et al. (2022), the default arguments are imported from the models respective GitHub directories. For NLinear and DLinear DARTS has been used. Lastly for RF the RandomForestRegressor in SKLearn and been used

Informer is somewhat competitive, especially when evaluated by MAD. It provides strong short-horizon accuracy and maintains near-best performance across several horizons, though it does not match the recursive linear neural networks beyond the short term. This suggests that attention-based models can capture temporal dependencies reasonably well.

Finally Autoformer and FEDFormer do not achieve top performance under either metric. Despite their flexibility and architectural complexity, they are consistently outperformed by simpler alternatives. The benchmark random walk performs worst overall, confirming that the models considered here extract meaningful predictive structure from the data.

Taken together, the results demonstrates the classic result that model complexity does not guarantee superior performance in global inflation forecasting. Recursive linear neural networks, which are both relatively simple to implement and way more computationally efficient than both RF and the transformer models, emerge as the most reliable choice across horizons. Also this brings an interesting twist to the claims made by Zeng et al. (2023), NLinear and DLinear does seem to outperform the transformer models (with Informer being the only somewhat competitive models), but with the addition the recursive forecasting strategy they perform even better. This indicates that it is not necessarily because of the recursive methods used to benchmark against transformer models they perform well, it is rather too simple methods they are benchmarked against. More sophisticated deep learning architectures, while theoretically appealing, do not outperform these simpler baselines in this empirical setting.

A somewhat different picture emerges from the summary statistics in Table 11. Whereas the recursive linear neural networks deliver the lowest average errors across horizons (Table 10), the Random Forest attains the largest share of individual country,horizon wins. For instance, RF accounts for roughly one-third of all best forecasts under both RMSE and MAD, and the aSPA test rejects the null of equal predictive ability against the random walk in more than 90% of countries. This suggests that RF is highly competitive on a case-by-case basis, even if its overall errors are not as low on average as recursive NLinear. By contrast, NLinear recursive achieves fewer outright wins but still performs strongly in the aSPA test. Also note that, yet again, the transformer models show a poor performance. Autoformer and FEDFormer achieve only a small share of individual wins and do not stand out in the aSPA test.

Table 2: Forecast accuracy comparison: RMSE and MAD by horizon. Minima are highlighted.

RMSE by Horizon

Model	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
Autoformer	0.813	0.773	0.764	0.752	0.736	0.723	0.726	$\underline{0.729}$	$\underline{0.726}$	0.741	0.802	0.797
FEDFormer	0.874	0.840	0.838	0.807	0.788	0.771	0.768	0.760	0.773	0.801	0.848	0.858
Informer	$\underline{0.720}$	$\underline{0.725}$	0.738	**0.719**	$\underline{0.730}$	0.734	$\underline{0.722}$	0.731	0.735	$\underline{0.720}$	0.721	0.737
RF	$\boxed{0.678}$	**0.723**	**0.733**	0.759	0.748	**0.690**	0.738	0.754	0.737	0.722	$\underline{0.691}$	$\boxed{0.574}$
DLinear	0.746	0.761	0.769	0.766	0.763	0.763	0.765	0.765	0.760	0.758	0.759	0.762
NLinear	0.722	0.733	$\underline{0.736}$	0.734	0.741	0.735	0.731	0.730	0.729	0.727	0.722	0.724
DLinear recursive	0.736	0.752	0.740	$\underline{0.724}$	**0.705**	$\underline{0.696}$	**0.693**	**0.684**	**0.678**	**0.677**	**0.670**	$\underline{0.664}$
NLinear recursive	**0.700**	$\boxed{0.716}$	$\boxed{0.703}$	$\boxed{0.685}$	$\boxed{0.665}$	$\boxed{0.658}$	$\boxed{0.656}$	$\boxed{0.648}$	$\boxed{0.643}$	$\boxed{0.640}$	$\boxed{0.634}$	**0.630**
RW	1.660	1.760	1.796	1.817	1.808	1.770	1.844	1.873	1.842	1.845	1.794	1.657

MAD by Horizon

Model	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
Autoformer	0.482	0.451	0.449	0.435	0.423	0.416	0.419	0.421	0.414	0.430	0.474	0.474
FEDFormer	0.529	0.501	0.487	0.482	0.469	0.450	0.454	0.450	0.450	0.468	0.505	0.507
Informer	**0.398**	**0.398**	$\boxed{0.408}$	$\boxed{0.387}$	**0.399**	$\underline{0.396}$	**0.395**	**0.397**	$\underline{0.397}$	0.392	0.391	0.404
RF	$\boxed{0.357}$	$\boxed{0.379}$	$\underline{0.416}$	$\underline{0.421}$	0.415	**0.393**	0.420	0.420	0.415	**0.384**	**0.360**	$\boxed{0.288}$
DLinear	0.444	0.461	0.455	0.463	0.458	0.468	0.460	0.461	0.449	0.448	0.454	0.453
NLinear	0.425	0.431	0.446	0.445	0.450	0.447	0.438	0.446	0.433	0.433	0.435	0.431
DLinear recursive	0.456	0.461	0.438	0.430	$\underline{0.414}$	0.409	$\underline{0.408}$	$\underline{0.400}$	**0.393**	$\underline{0.388}$	$\underline{0.384}$	$\underline{0.372}$
NLinear recursive	$\underline{0.408}$	$\underline{0.420}$	**0.414**	**0.396**	$\boxed{0.385}$	$\boxed{0.374}$	$\boxed{0.372}$	$\boxed{0.368}$	$\boxed{0.367}$	$\boxed{0.359}$	$\boxed{0.356}$	**0.353**
RW	0.578	0.660	0.695	0.718	0.708	0.684	0.711	0.730	0.719	0.697	0.645	0.533

The table reports average forecast errors across all countries for horizons $h = 1,\dots,12$. Performance is measured by Root Mean Squared Error (RMSE) and Median Absolute Deviation (MAD). Boxed values are minima across the horizon, bold values are second smallest and underscored values are third smallest.

Informer performs somewhat better, with a reasonable share of wins and rejection rates, but still falls short of recursive NLinear and RF. This underlines once again that while transformer architectures may capture temporal dependencies, their added complexity does not translate into systematic forecasting gains in this setting. Together, these results highlight an important distinction between the average accuracy and the frequency of wins. RF appears well suited to exploiting nonlinearities in specific series or horizons, but recursive NLinear remain more consistent performers across the global panel. This shows the importance of model selection; complex nonlinear models can be locally useful in some cases, while a linear model might be preferable in others.

Summary Statistics				
Model	**Freq_RMSE**	**Freq_MAD**	**aSPA_mean**	**aSPA_share**
Autoformer	0.053	0.048	0.152	**0.767**
FEDFormer	0.018	0.028	0.214	0.644
Informer	0.075	0.097	0.174	0.733
RF	0.345	0.377	0.044	0.921
DLinear	0.027	0.023	0.303	0.533
NLinear	0.068	0.051	0.231	0.611
DLinear recursive	0.047	0.073	0.192	0.733
NLinear recursive	**0.250**	**0.187**	**0.145**	**0.767**
RW	0.001	0.006		

Table 3: Summary statistics across models.

Freq_RMSE and Freq_MAD show the share of country–horizon pairs where a model achieves the lowest error. aSPA_mean reports the mean p-value of the aSPA test against the random walk benchmark, while aSPA_share gives the fraction of countries where the null of equal predictive ability is rejected at the 5% level.

While multi-step forecasts of monthly inflation rates provide information about short-term predictive accuracy, an equally important question is whether models can capture the evolution of inflation over longer horizons. Cumulative inflation is of particular relevance for policy analysis and economic decision-making, since central banks typically focus on inflation developments over annual or multi-quarter horizons rather than on individual monthly outcomes.[40, 24]

Table 4: Cumulative inflation forecast errors

Cumulative RMSE by Horizon

Model	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
Autoformer	0.813	1.294	1.764	2.210	2.576	2.879	3.227	3.610	3.971	4.291	4.545	4.719
FEDFormer	0.874	1.368	1.848	2.323	2.719	3.063	3.465	3.897	4.295	4.657	4.946	5.100
Informer	0.720	**1.149**	**1.555**	**1.910**	**2.266**	**2.600**	**2.959**	**3.335**	**3.685**	**3.984**	**4.227**	**4.410**
RF	0.678	0.966	1.191	1.399	1.569	1.707	1.908	2.084	2.249	2.421	2.591	2.736
DLinear	0.746	1.329	1.897	2.441	3.011	3.548	4.027	4.558	5.125	5.630	6.112	6.532
NLinear	0.722	1.302	1.870	2.429	3.007	3.539	4.008	4.539	5.092	5.584	6.057	6.470
DLinear recursive	0.736	1.332	1.904	2.434	2.898	3.322	3.680	4.108	4.569	4.998	5.396	5.727
NLinear recursive	**0.700**	1.266	1.800	2.288	2.730	3.135	3.474	3.883	4.323	4.734	5.109	5.420
RW	1.660	2.573	3.316	3.875	4.396	4.885	5.294	5.680	6.032	6.385	6.788	7.241

Cumulative MAD by Horizon

Model	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
Autoformer	0.482	0.821	1.155	1.434	1.681	1.874	2.103	2.362	2.626	2.796	2.985	3.094
FEDFormer	0.529	0.875	1.218	1.530	1.795	2.063	2.301	2.548	2.843	3.058	3.224	3.278
Informer	**0.398**	**0.669**	**0.933**	**1.134**	**1.335**	**1.534**	**1.735**	**1.946**	**2.118**	**2.262**	**2.349**	**2.389**
RF	0.357	0.537	0.687	0.788	0.877	0.926	1.014	1.096	1.177	1.221	1.276	1.291
DLinear	0.444	0.823	1.227	1.593	1.927	2.244	2.552	2.895	3.218	3.566	3.835	4.060
NLinear	0.425	0.810	1.206	1.584	1.941	2.229	2.535	2.863	3.256	3.532	3.792	3.996
DLinear recursive	0.456	0.832	1.234	1.604	1.905	2.137	2.334	2.616	2.893	3.136	3.383	3.447
NLinear recursive	0.408	0.781	1.147	1.496	1.761	2.003	2.210	2.449	2.738	2.982	3.128	3.318
RW	0.578	1.029	1.427	1.725	1.970	2.133	2.234	2.296	2.353	2.367	2.407	2.523

Cumulative forecasts obtained by summing monthly inflation predictions up to horizon h, and evaluated against realized cumulative inflation.

Evaluating cumulative forecasts provides complementary insights to horizon-specific errors. A model that performs well at $h = 1$ may still accumulate systematic biases over time, while one with slightly higher short-run error can deliver more reliable medium-term projections if its errors are less persistent. Assessing cumulative forecast errors thus captures how well models track the build-up of inflationary pressure, a measure more relevant for monetary policy and macroeconomic planning. While the models have not been specifically trained for forecasting the cumulative inflation it is still reported here.

Formally, the cumulative h-step forecast of inflation at time t is defined as

$$\hat{\Pi}_{t,h} = \sum_{j=1}^{h} \hat{\pi}_{t+j|t},$$

where $\hat{\pi}_{t+j|t}$ denotes the forecast of monthly inflation j periods ahead made at time t. Table 4 shows that Random Forest clearly dominates when forecasts are evaluated cumulatively, with the lowest RMSE and MAD at every horizon. This indicates that RF is particularly effective in controlling error growth as forecasts are aggregated. Informer also performs decently well in the cumulative MAD, which show that smoother predictions translate into stable multi-month inflation paths. By contrast, the recursive linear models, despite their strength in horizon-specific accuracy, lose part of their advantage once errors are summed, as small deviations accumulate over time. Autoformer and FEDFormer again lag behind, offering little improvement in this evaluation. Overall, these findings highlight that model rankings depend on the evaluation metric: recursive linear neural networks are most reliable for individual horizons, while RF is especially competitive for cumulative inflation forecasts.

To complement these comparisons, Table 5 show the results of the aSPA test of the presented model vs. Random Forest. The reported values summarize, for each horizon and model, the mean p-value across countries as well as the share of countries in which the null hypothesis of equal predictive accuracy cannot be rejected at the 5% level.

At shorter (1–3 months ahead), the average p-values are high across models, and the share of countries where the null cannot be rejected remains limited. At intermediate horizons (4–10 months), Informer and recursive NLinear model consistently achieve the highest share, indicating that their forecasts is most often not signficantly from that of Random Forest across countries. In general all models' forecasts are never significantly better than RF. Autoformer and FEDFormer is still performing bad compared to the other models while Informer does seem to be reasonable for the early horizons.

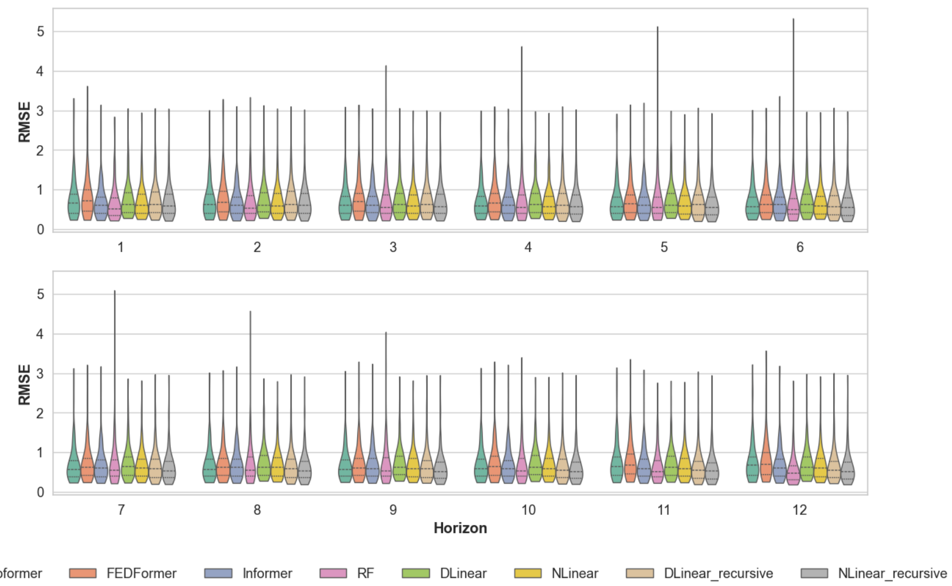
Table 5: aSPA test results of model vs. Random Forest across horizons.

model	Autoformer_h12		DLinear_h12		DLinear_rec		FEDFormer_h12		Informer_h12		NLinear_h12		NLinear_rec	
Horizon	Mean	Share	Mean	Share	Mean	Share	Mean	Share	Mean	Share	Mean	Share	Mean	Share
h = 1	0.718	0.133	0.696	0.178	0.644	0.222	0.798	0.078	**0.615**	0.267	0.647	0.222	0.589	**0.244**
h = 2	0.607	0.244	0.649	0.211	0.624	0.244	0.675	0.133	0.522	0.300	0.600	0.267	**0.562**	0.300
h = 3	0.541	0.267	0.633	0.211	0.569	0.289	0.635	0.178	0.502	0.333	0.572	0.244	**0.518**	**0.322**
h = 4	0.492	0.311	0.613	0.222	0.501	0.322	0.563	0.256	**0.446**	0.400	0.553	0.289	0.444	**0.367**
h = 5	0.481	0.300	0.627	0.222	**0.480**	**0.333**	0.582	0.233	0.483	**0.333**	0.581	0.222	0.397	0.367
h = 6	0.597	0.200	0.731	0.111	**0.578**	0.211	0.701	0.100	0.634	**0.222**	0.676	0.167	0.492	0.289
h = 7	0.480	0.289	0.663	0.200	**0.455**	**0.311**	0.568	0.200	0.463	**0.311**	0.597	0.222	0.368	0.367
h = 8	**0.420**	0.378	0.626	0.244	0.424	0.378	0.491	0.289	0.470	**0.433**	0.547	0.256	0.335	0.444
h = 9	**0.439**	**0.411**	0.630	0.189	0.440	**0.411**	0.556	0.222	0.484	0.367	0.564	0.244	0.347	0.433
h = 10	0.480	0.311	0.645	0.200	**0.450**	0.333	0.609	0.178	0.459	**0.367**	0.583	0.233	0.342	0.422
h = 11	0.682	0.144	0.683	0.189	**0.459**	0.356	0.755	0.056	0.554	0.267	0.644	0.211	0.372	0.356
h = 12	0.915	0.000	0.954	0.000	**0.798**	**0.022**	0.964	0.000	0.937	0.000	0.924	0.000	0.712	0.100

Mean reports the average aSPA statistic across countries for each model and horizon, while Share reports the proportion of countries where the model is not rejected at the 5% level.

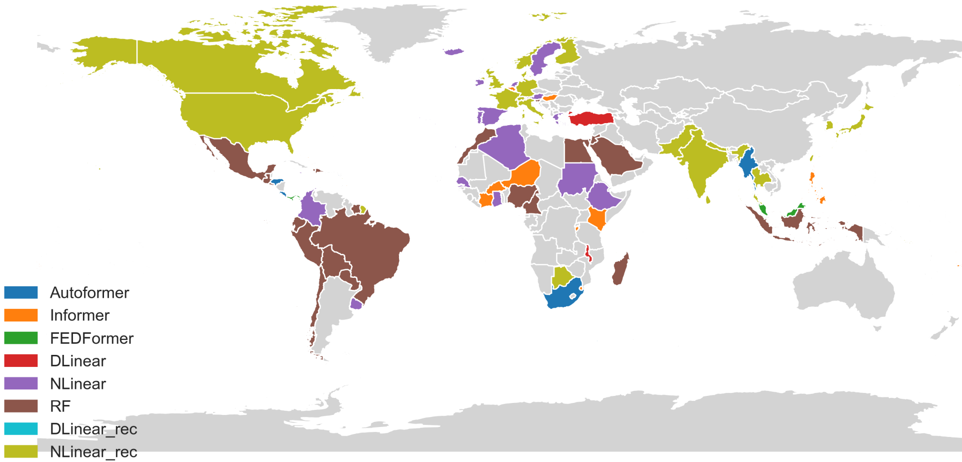
Up until now only mean aggregated results and predictability test have been shown. Figure 6 shows violin plots of RMSE statistics for all models across forecast horizons $h = 1$–12.

Figure 6: Distribution of RMSE across models and horizons



Violin plots show the distribution of RMSE across countries for each model and forecast horizon. Inner lines mark quartiles, and horizontal width reflects density. A shared legend is placed below the panels.

Figure 7: Best-performing forecasting model by country



This shows the best-performing forecasting model for inflation in each country, determined by the lowest average RMSE across prediction horizons. Countries are shaded according to the model with the lowest mean RMSE, with distinct colors representing.

From the above above results it seems that at intermediate horizons (4–10 months), Informer and recursiveNLinear perform well (especially recursive). By contrast, models such as FEDFormer, Autoformer, and DLinear occasionally match these models.. At the 12-month horizon RF is the consistently best performer as well as for the cumulative inflation. This reinforces the claims made by Medeiros et al. (2021), Medeiros, Schütte, and Soussi (2023)

These observations are reinforced by the distribution of forecast errors shown in 5. Errors are pretty tightly clustered for all models, interestingly some pretty extreme outliers can be seen in the case of RF, this could be the reason for not best performance in the aggregate results, while still performing great in the aSPA tests. But in general the same picture is seen across all of the horizons, random forests mean is quite low but does seem more narrow than the other models. Generally the models error distributions looks quite comparable from this plot at least.

Figure 7 displays the best-performing forecasting model in each country, based on the lowest mean RMSE across prediction horizons. The map looks quite diverse. In Latin America, RF is strong (brown) dominates, covering most of South America and parts of Central America, with only a few exceptions in the lower Central America where NLinear, Autoformer and FEDFormer have some wins. Africa is quite diverse. In sub-Saharan Africa RF and NLinear (purple) appear frequently, while Informer (orange) performs better in some West and Central African countries. North America, India, and several East Asian economies are characterized by recursive NLinear (yellow-green), suggesting that recursive linear structures capture their inflation dynamics relatively well. Turkey stands out as a rare case where DLinear (red) outperforms all other approaches. Much of Europe shows a patchwork, with NLinear and Random Forest both taking prominent roles. Notably recursive DLinear does not win in any countries. The results a summarised in table 6

Model	country wins
RF	26
NLinear_rec	21
NLinear	20
Informer	13
Autoformer	7
DLinear	2
FEDFormer	2

Table 6: Number of countries for which each forecasting model achieves the lowest mean RMSE across horizons.

These regional differences are in line with the arguments by Ciccarelli and Mojon (2010) who show that a significant portion of national inflation dynamics can be explained by a common global factor, which could help to interpret why transformer-based models that exploit nonlinear shared dynamics succeed in some countries. This could explain why Random Forest emerges as the best model in many countries across the region. In such contexts, relatively simple models may perform well because it is only only lagging itself while still getting the current inflation for other countries.

For Africa Franses and Janssens (2018) argues for heterogeneity in inflation dynamics across African countries. They show that the inflation in Africa has more differences than common features, making it inappropriate to speak of an “average inflation in Africa.” Instead, they argue that models should be constructed country by country. This is exactly what is seen here a diverse pattern of best-performing models observed in Figure **??**, where no single forecasting approach dominates the region.

Ha, Kose, and Ohnsorge (2019) show that inflation has become increasingly influence by global dynamics since the 2000s. At the same time, they argue that regional and domestic influences remain important, especially in emerging markets, where inflation is still partly driven by local conditions in addition to global forces. The figure reflects this complexity by showing the diversity of the best performing model.

Thus while the figure show model accuracy by country it does, while being purely speculative, show how diverse inflation on a global scale is.

A similar analysis has also been done for each horizon. The results can be seen in Table 7 and plot like figure 7 has been done in E in the appendix

Horizon	1	2	3	4	5	6	7	8	9	10	11	12	Total
Autoformer	5	5	7	6	7	7	6	9	10	6	2	1	71
DLinear	4	6	3	3	1	3	4	5	3	2	1	0	35
DLinear rec.	2	0	0	1	1	2	5	5	10	11	10	4	51
FEDFormer	2	1	1	1	4	1	4	3	2	1	2	0	22
Informer	9	16	17	18	13	8	9	5	6	10	11	3	125
NLinear	18	17	20	17	15	9	10	9	10	6	6	0	137
NLinear rec.	8	9	15	19	24	23	**29**	**30**	**28**	**34**	**36**	18	273
RF	**43**	**37**	**28**	**26**	**26**	**38**	24	25	22	21	23	**64**	**377**
RW	0	0	0	0	0	0	0	0	0	0	0	1	1

Table 7: Number of countries for which each forecasting model achieves the lowest mean RMSE across countries at different horizons.

RF is the most successful model overall, with 377 wins, and it dominates particularly at short horizons. Its relative advantage a bit at longer horizons, where recursive NLinear takes over, and then a sudden shift at $h = 12$ with a clear advantage for RF.

Especically the recursive NLinear improves steadily with the horizon, overtaking RF from $h = 7$ to $h = 11$ and achieving the second-highest total with 273 wins. This is surpricing as one would expect that performance declines as the horizon increases, again this is similar to the result in Marcellino, Stock, and Watson (2006). Interestingly it is opposite for NLinear. Once again the transformers show disappointing results, once again it can be seen that Informer takes the cake for the best transformer model, while FEDFormer is the worst. An intersting observation is that Informer starts doing okay at horizon two and keeps increasing until $h = 12$. Random walk is winning only once at $h = 12$ in the Netherlands.

Looking at the figures in section E. There are clear shifts in model performance emerge. At short horizons ($h = 1$–$h = 4$), results are more heterogeneous, Transformer-based models such as Informer and Autoformer perform well in parts of North America, Asia, and selected African countries, while RF dominates much of Latin America and some parts of Africa. As the forecast horizon lengthens, recursive NLinear, gain prominence, especially in North America, Europe, and South Asia. By the longest horizons ($h = 9$–$h = 12$), the picture converges, with RF and recursive NLinear being the winning approaches across nearly all regions. Similar arguments could be made for these plots as done for Figure 7.

For showing the performance across time the cumulative RMSE reported in Figure 8 is computed in three steps. First, for each country i, horizon h, and evaluation period t, the squared forecast error is defined as

$$e^2_{i,h,t} = \left(y_{i,t} - \hat{y}_{i,h,t}\right)^2,$$

where $y_{i,t}$ denotes the observed inflation and $\hat{y}_{i,h,t}$ the h-step-ahead forecast.

Second, the squared errors are averaged across horizons and countries with equal weights to obtain a panel-wide mean squared error at time t:

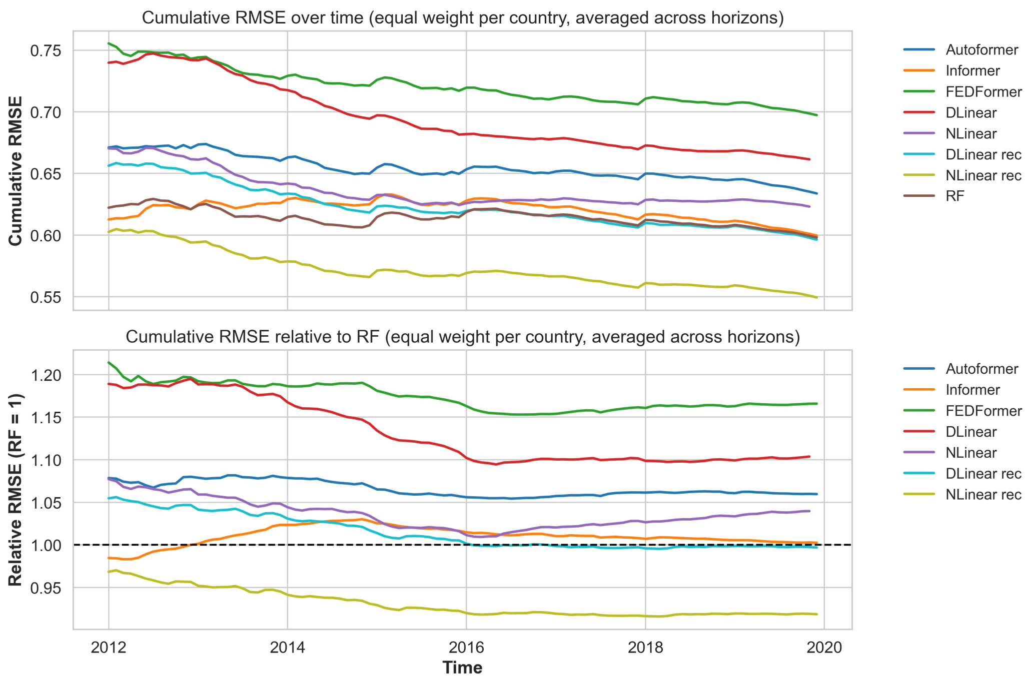
$$\overline{e^2_t} = \frac{1}{NH} \sum_{i=1}^{N} \sum_{h=1}^{H} e^2_{i,h,t},$$

where N is the number of countries and H the number of horizons.

Finally, the cumulative RMSE up to period T is defined as the expanding root mean squared error:

$$\text{CRMSE}_T = \sqrt{\frac{1}{T} \sum_{t=1}^{T} \overline{e^2_t}} \tag{19}$$

Figure 8: Cumulative RMSE performance of forecasting models



This figure plots the cumulative root mean squared error (RMSE) of each forecasting model over time. Errors are computed as in equation 19

Looking at Figure 8 the transformer-based architectures (Autoformer, Informer, and FEDFormer), once again, fail to match the accuracy of simpler benchmarks. FEDFormer stands out as the consistently weakest performer. Informer starts out strong but only manages a marginal improvement toward the end of the evaluation period. Autoformer, meanwhile, never surpasses RF in relative terms.

For the linear neural networks the picture is more nuanced. The recurrently trained variants are the strong performer. Recursive DLinear tracks RF closely but shows barely any improvement at the end, whereas recursive NLinear consistently outperforms the other models. In fact, Recursive NLinear steadily diverges from the rest of the models, ultimately delivering around 6-10% lower cumulative RMSE compared to RF at the end of the evaluation period. Meanwhile DLinear is the second worst performer across the entire period, and NLinear, is starting of bad, coming closer and ends up about six percent worse than RF.

Thus while transformer-based models struggle to deliver gains in this forecasting environment with only informer coming close, the recursive linear neural networks, especially recursive NLinear, is are consistent across time.

4.3 Variable Importance

Up until now from the forecast results it seems that recursive NLinear and RF, emerge as the most reliable performers, while transformer-based models struggle to obtain strong forecasts. These accuracy comparisons only show how the models perform, no light has been shed on why the differences arise. The analysis so far establishes relative predictive performance but remains silent on the mechanisms driving it. Understanding the underlying drivers of forecast accuracy is crucial especially for these so called black-box methods. Both to assess whether models exploit the economic structure or merely statistical patterns, and to evaluate their usefulness for policy analysis. To this end, this section turns to an examination of variable importance. By employing SHAP values described in section 3.8, the estimated importance of individual predictors to the forecasts can be shown, offering a decomposition of model outputs to more interpretable components. The aim is to provide a complementary dimension to the error-based evaluation, helping to explain why models differ in performance and what variables matter most in driving the inflation forecasts. For all models with a horizon length of 12 two plots have been created.

First the bar plots have been created with the following method.

Let $\phi_{i,t,h}$ denote the SHAP value of predictor i at forecast origin t and horizon h. The variable importance for predictor group g at horizon h is defined as

$$VI_{g,h} = \sum_{i \in g} \frac{1}{T} \sum_{t=1}^{T} |\phi_{i,t,h}|, \tag{20}$$

where T is the number of steps in the rolling window. For comparability across all continents, the values are normalized within each target continent:

$$\widetilde{VI}_{g,h} = \frac{VI_{g,h}}{\sum_{g'} VI_{g',h}}. \tag{21}$$

Note that the predictor group definition can vary across models, this will be introduced later. Using this a stacked bar plot is created with $\widetilde{VI}_{g,h}$ across predictor groups, horizons, and continents. This plots highlight which blocks of predictors drive the forecasts at different horizons and in different continents.

Next a heatmap for the lags. To compliment the bar plot a heatmap of the VI is compressed along the lag dimension. Here $\phi_{i,\ell,t,h}$ denote the SHAP value for lag ℓ of predictor i at forecast origin t and horizon h. The aggragtion here is

$$VI_{\ell,h} = \sum_i \frac{1}{T} \sum_{t=1}^{T} |\phi_{i,\ell,t,h}|, \tag{22}$$

And with the following normalization

$$\widetilde{VI}_{\ell,h} = \frac{VI_{\ell,h}}{\sum_{\ell,h} VI_{\ell,h}}. \tag{23}$$

Heatmaps of $\widetilde{VI}_{\ell,h}$ reveal which lags matter most for short-term versus long-term horizons.

Note that the amount of lags have been tuned across models, because of this there are some forecasts that does not contain the same of lags as others withing the same model. This is handled by setting the shap value for the non-included to 0 as it does not contribute with anything to the forecast. While this does not show the real importance of the lag when it is included it is simply a way of showing how the training procedure weights variables.

For RF the predictor groups are the global inflation factor, autoregressive lags of only the target (AR), continent-specific inflation series (Africa, Asia & Oceania, Europe, North America, South America), and monthly dummies.

Looking at figure 9 the autoregressive (AR) lags have the largest importance across all horizons and continents. This shows the strong persistence in inflation and suggests that RF primarily exploits the autoregressive structure to generate forecasts, this also argues for the choice of tuning the amount of lags as a hyperparameter instead of just fixing. Inflation from other continents, contribute to varying degrees depending on the target region, however, it does seem that at least half of the variable importance across horizons comes from other countries that the target. Especially Europe show a large influence by the AR. Monthly dummies contribute only modestly, indicating that RF relies more heavily on the AR to catch seasonality, but they are still visible, unlike the global factor appears negligible at almost all horizons, but does pop up in North- and South America.

For Africa, Asia & Oceania, and Europe, the heatmaps exhibit a pronounced diagonal structure, where lag $t-h$ contributes most strongly to the prediction at horizon h. This pattern is consistent with an autoregressive propagation of shocks, and also shows that RF succesfully finds thess. Importantly, the intensity of SHAP values decays as the horizon increases, suggesting that the predictive relevance of past inflation gets smaller over time. These regions therefore exhibit dynamics similar to a standard autoregressive process, this also shows the importance of the autoregressive variables and model as argud by [40, 42]. The patterns for North America and South America looks more messy. In North America, the first lag dominates across nearly all horizons, with far less emphasis on the diagonal structure. This indicates that inflation seems to be driven by the most recent observation, and that the RF put a large weight of the forecast path forward from this single input. South America shows an intermediate pattern: while the first lag is still quite strong, there is some evidence of diagonal propagation, albeit weaker than in Africa, Asia & Oceania, and Europe.

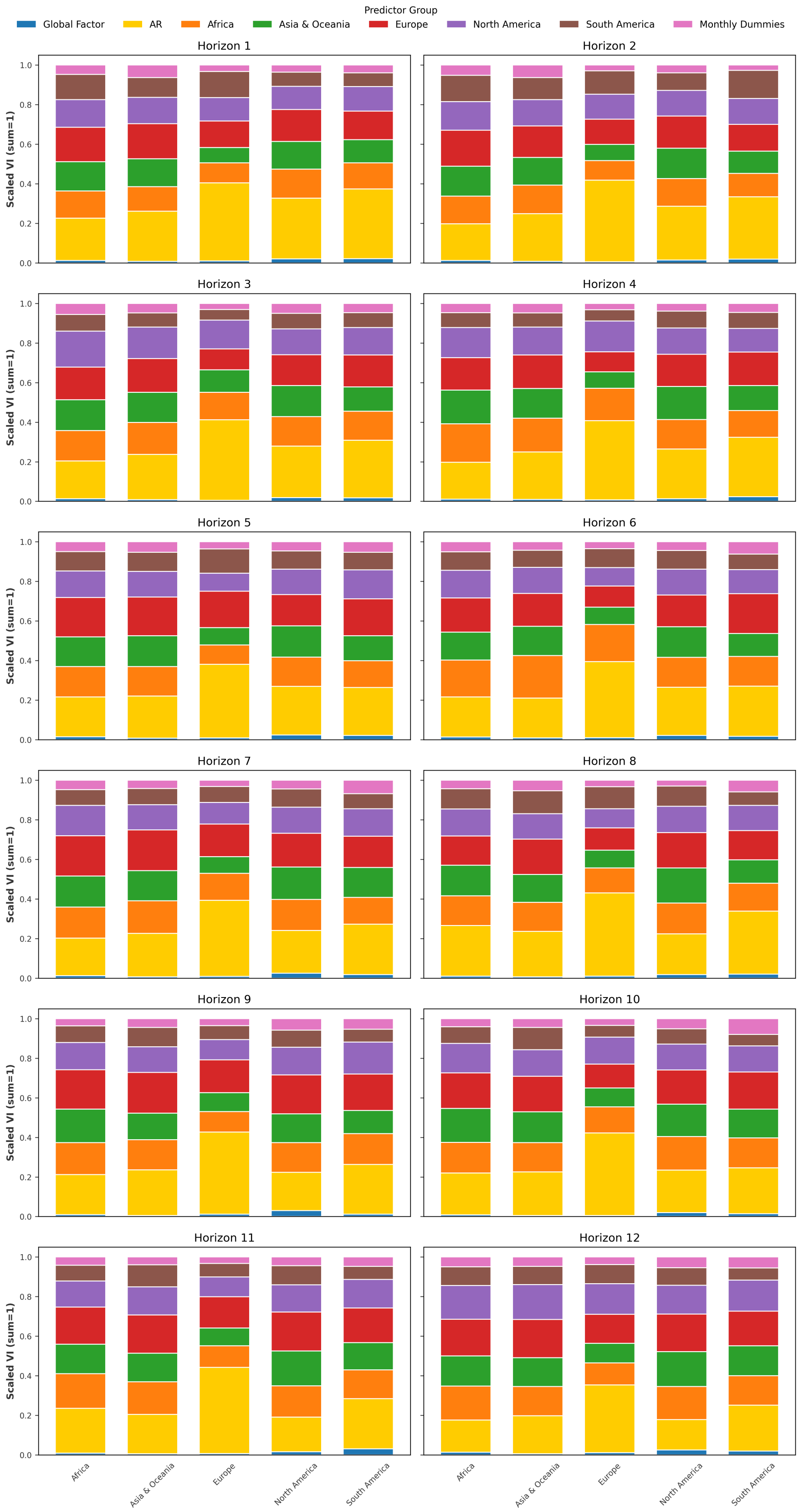


Figure 9: Random Forest: Predictor groups across horizons and continents.

For Informer the predictor groups are the global inflation factor, continent-

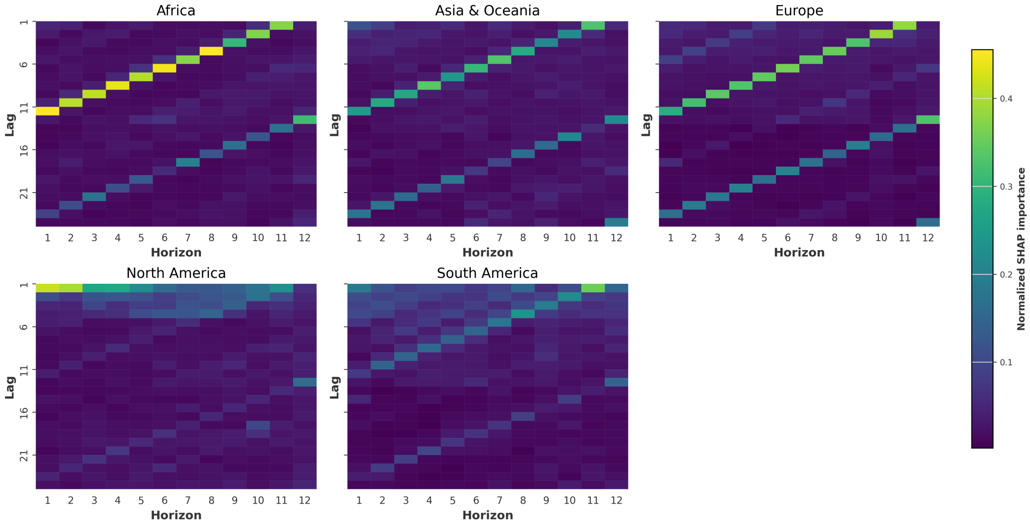


Figure 10: Random Forest: Lag–horizon heatmap of variable importance.

specific inflation series (Africa, Asia & Oceania, Europe, North America, South America), and monthly dummies. Unlike RF, Informer does not use only the lags as of the target country since the model has multidimensional target, but instead recieves lags of all countries and global factor at once, furthermore the there are no monthly dummies but rather the time encoding (TE).

Looking at Figure 12, it does seem that generally for most continents the continent itself has the largest importance. Though this is more pronounced for Europe than for other continents, but generally there seems to be a somewhat evenly split among most continents, interestingly the TE does not have a noticable impact on most of the continents and horizons. The big exceptio to all of this is South America. Here, for some reason, Africa has a huge influence, TE and global factor shows up here as well. This is not seen in the results by RF. And with keeping in mind that Informer, while being the best transformer model, did not do great, it seems that the model had a hard time handling the hyperinflation in South America and then tried to use other measure to estimate it. Another observation is that the importance is quite stable across horizons, this is similar to what has been seen if Figure 9.

The lag–horizon SHAP heatmaps in Figure 11 show a very different pattern compared to RF. The diagonal structure is completely gone. Informer seems to care most about the most recent lags (1–3), but but with clear lines at seasonal intervals (around lag 12, 24, and 36). This clearly shows that it is not modelled with one model for each horizon, but one global model. This indicates that the ProbSparse attention used in Informer might be too sparse. As argued by Giannone, Lenza, and Primiceri (2021) macro-economic series are often not sparse but many predictors often lead to a better results. So here it seems that by using sparsity Informer has just made it harder for it self.

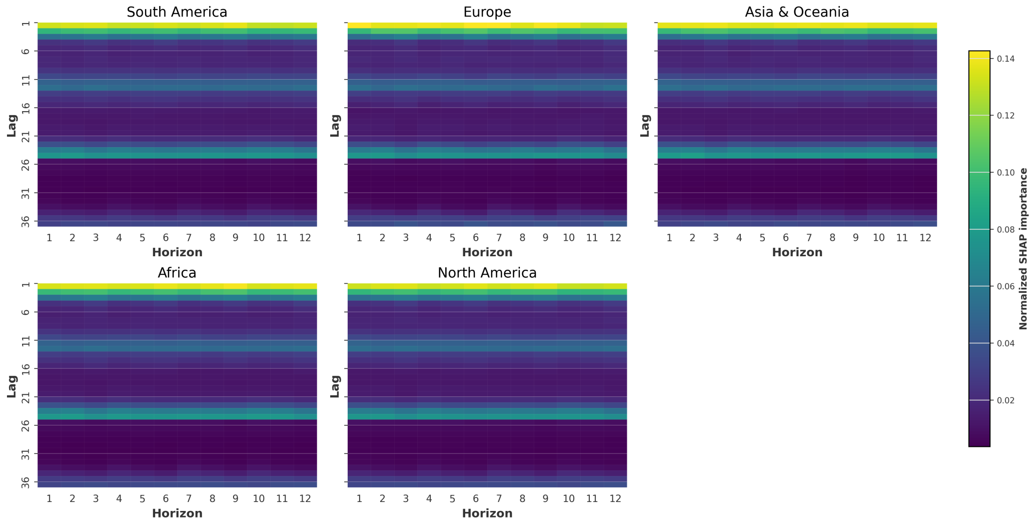


Figure 11: Informer: Lag–horizon heatmap of variable importance.

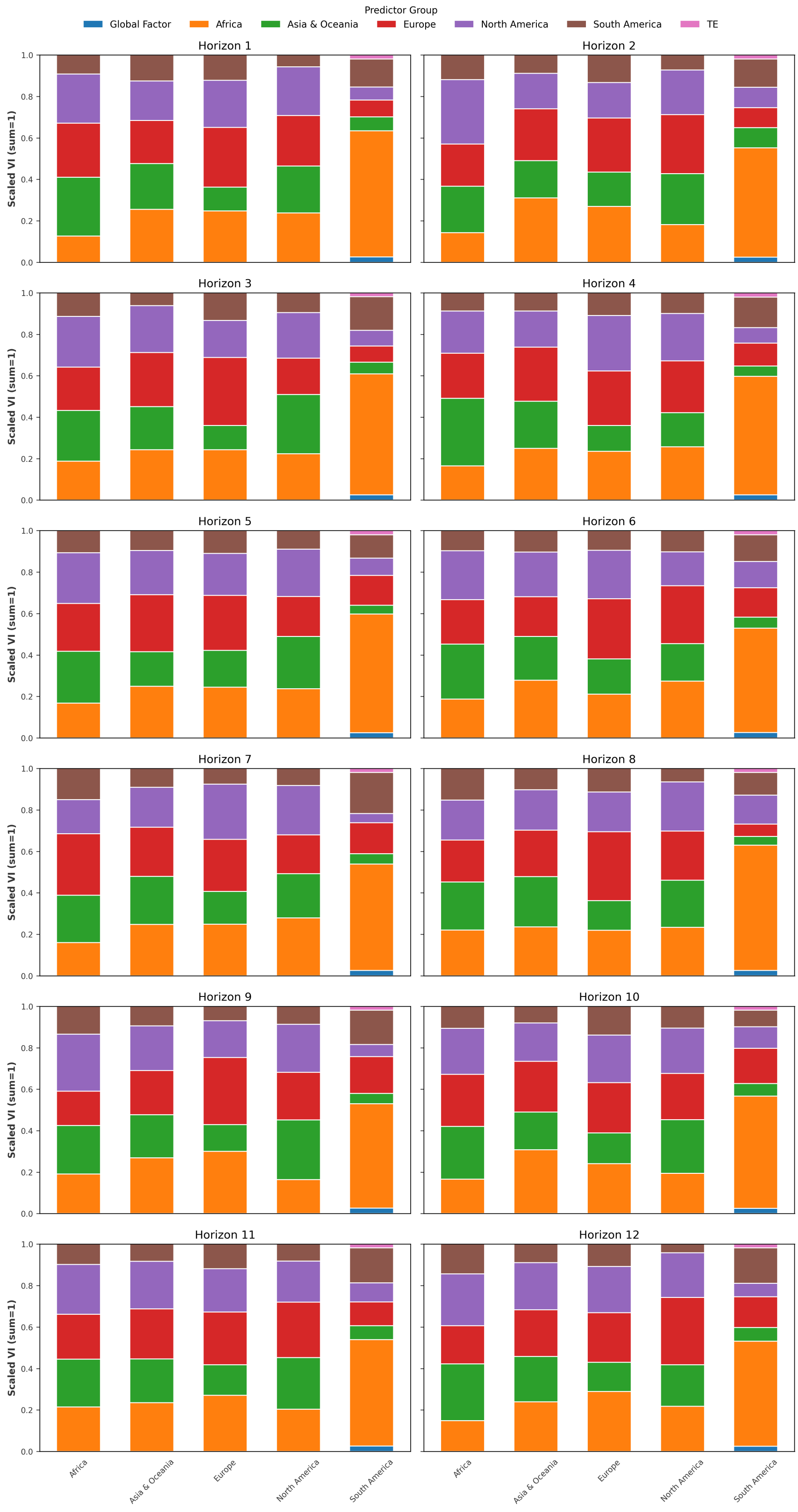


Figure 12: Informer: Predictor groups across horizons and continents.

Moving on to the second worst transformer model: Autoformer. Here the predictor groups are exactly the same as in Informer.

From Figure 13 its seems that Autoformer distributes more evenly across continents, but unlike Informer, the contributions vary a lot across horizons. The greater variance in contribution could indicate that Autoformer's autocorrelation mechanism shifts which inputs it emphasizes depending on how far ahead the forecast is. While it is hard to say if this is exactly the it does match the idea from Autoformer that the autocorrelation attention is built to capture periodic and long-term dependencies. Keeping in mind that the performance of Autoformer is bad it is hard to say if it is exactly the autocorrelation focus that is the error. However as it was for Informer South America again stands out with a different behavior, still Africa is the mos dominant and it is the continent where TE and the Global factor have a noticable impact.

From the heatmap shown in figure 14 the expected diagonal is, once again, not seen the expected diagonal. But a pattern does appear, for the first horizon there is emphasis on the lag $t-12$, and for the last horizon there is a large emphasis huge emphasis on lag 1 and then $13, 25$. And towards the middle of the plot it becomes blurry. From this plot it does not seem that the autocorrelation attention captures periodic and long-term dependencies, at least not in a clear way. Also the importance looks basically constant, which is curious. The clear difference that was seen in RF is gone here. The same applies to Informer. Taking into account the poor performance of Autoformer and DLinear it does not seem that the seasonal decomposition and autocorrelation attention provides a meaningful performance boost for inflation forecasting.

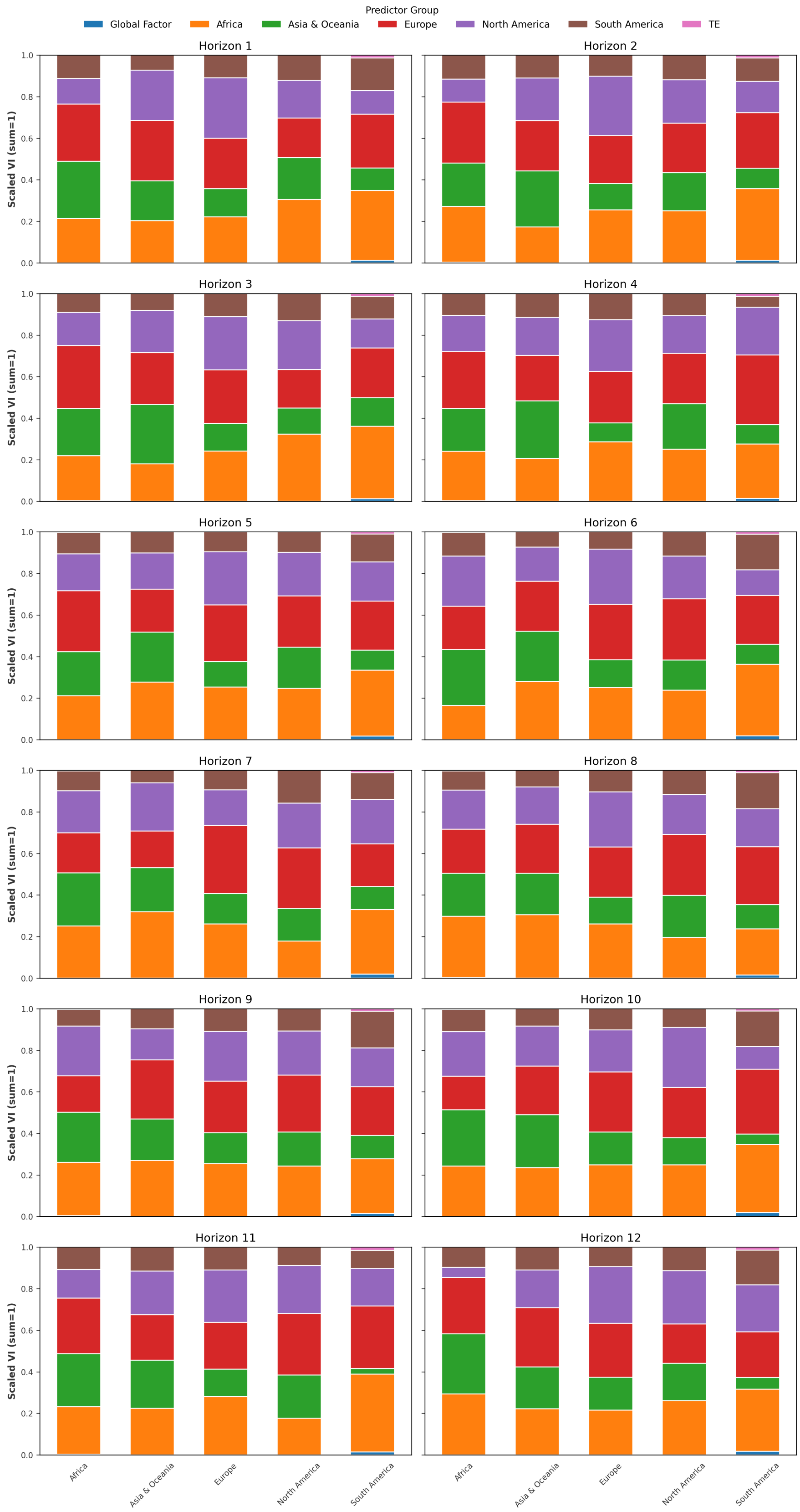


Figure 13: Autoformer: Predictor groups across horizons and continents.

FEDFormer extends the decomposition idea by relying more heavily on

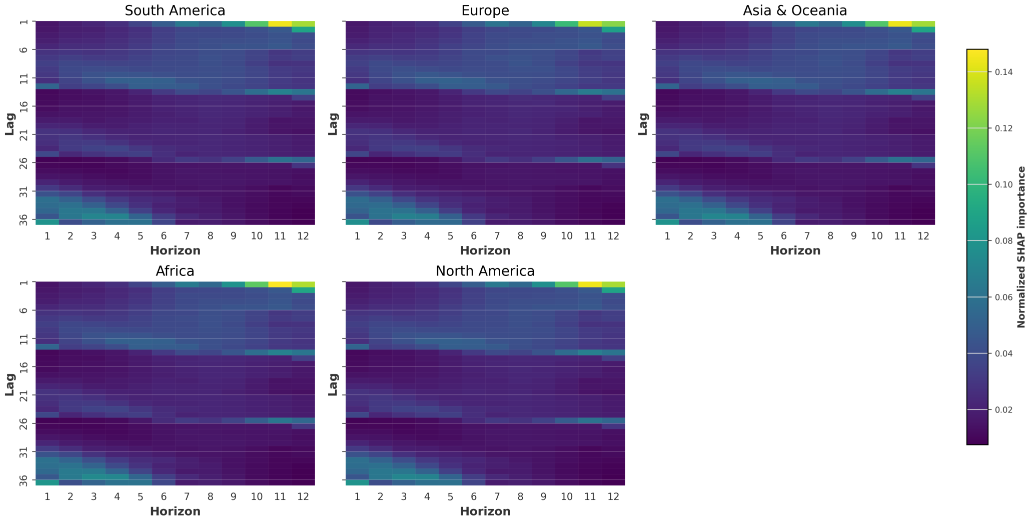


Figure 14: Autoformer: Lag–horizon heatmap of variable importance.

the frequency domain. it projects sequences to Fourier space and concentrates on a small set of dominant frequencies, then reconstructs in the time domain. Conceptually, this biases the model toward persistent seasonal cycles and smooth components—akin to Autoformer's autocorrelation attention, but implemented via spectral selection rather than time-domain autocorrelation. The variable importance plots can be found in appendix F FEDFormer looks to act in a similar way as Autoformer. Importance is distributed across regions with horizon-dependent shifts FEDFormer seems slightly more stable, but only slightly. The heatmap looks almost identican to Autoformer, suggesting that a frequency-focused approach is not sufficient to handle inflation's structural breaks and regime shifts in our setting.

Moving on from the transformer model and taking a look at recursive NLinear. NLinear in the recursive setup uses the global factor, monthly dummies, and continent-specific inflation series as predictors. Figure 15 show a mostly even allocation across continents for all horizons, with the global factor having very little impact and the largest contribution from monthly dummies this far. The distribution is very stable across horizons, indicating that the recursive one-step-ahead mechanism reuses a similar mix of signals as h increases, which of course does make sense, since it is the same model with no update in training making each forecast.

The heatmap in Figure 16 shows that the smallest lag length (4) is chosen by every single hyperparameter tuning, suggesting that importance concentrates on a stable subset of lags. This compliments Marcellino, Stock, and Watson (2006), who argues that short-lag iterated models often perform best in recursive settings for macroeconomic series. Surprisingly most of the weight is put on the 4th lag. A likely cause for this is the normalization done by NLinear since subtracts the latest observation from the input series (with a length of 4) in this case it is likely that. The most clear differences is likely in lag 4. The next question is then why is the first row not 0, since when training all values for the first lag is zero. And the answer is that the normalization also has an impact in fact looking at NLinear results in the appendix F it seems that the more lags is used the more important the first lag is.

In summary while RF leans on a clear seasonal patterns recursive NLinear keeps persistence in a different way. The last observation normalization smooths the signal instead of pushing shocks step by step. And focuses on very recent data. Informer focuses too narrowly on a few lags (1–3 and 12/24/36), so it can miss useful mid-range lags. Autoformer shifts what it uses by horizon and leans on seasonality; that can be brittle when patterns drift. Also this shows the importance of proper preprocessing of data, with just a very simple normalization of the data to handle distribution shifts it is performing much better than any of

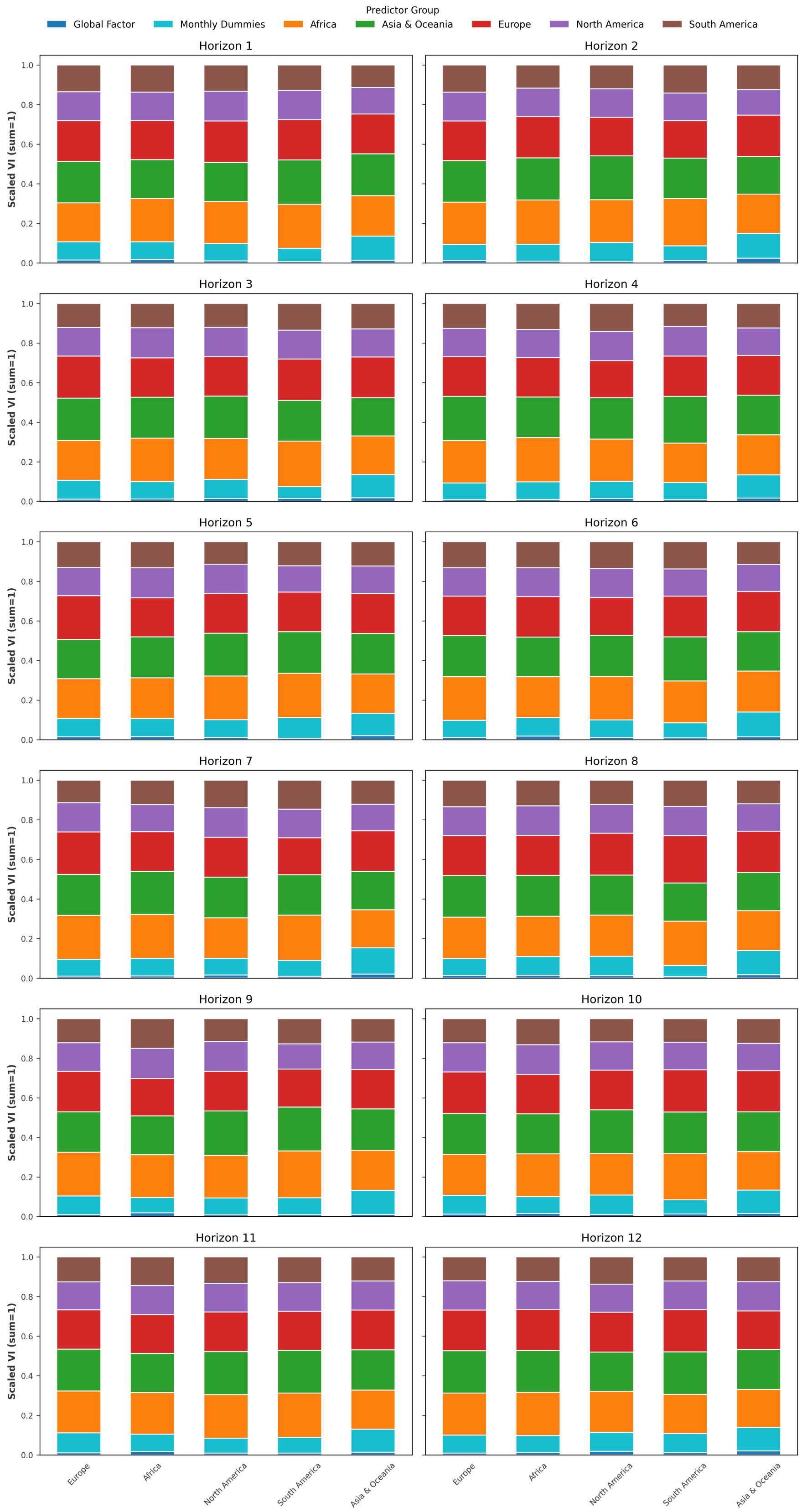


Figure 15: Recursive NLinear: Predictor groups across horizons and continents.

All interpretations above are based on estimated(!) SHAP values and

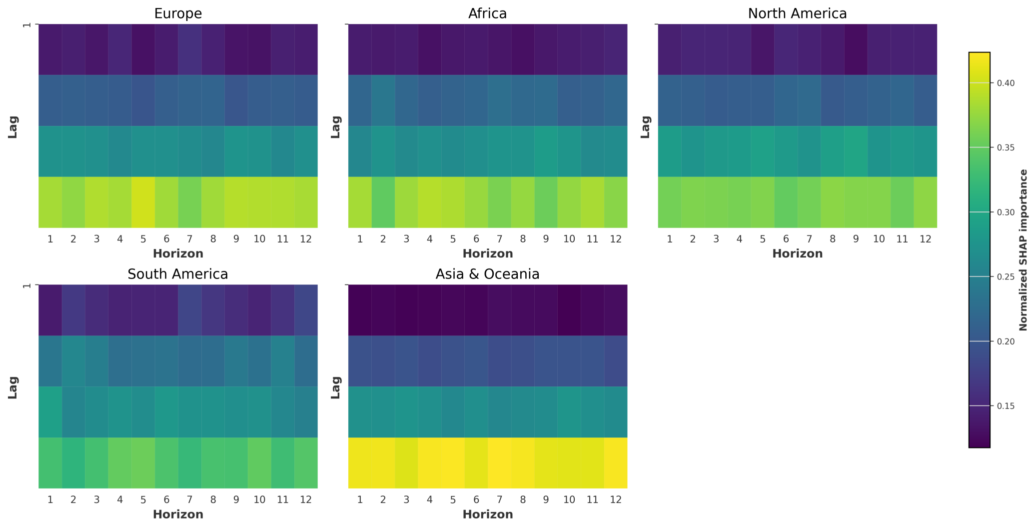


Figure 16: NLinear: Lag horizon heatmap of variable importance.

the chosen normalizations/tuning and handling of missing lags. No extra robustness checks (e.g., alternative normalizations, seeds, or ablations). The VI therefore shows what these trained models used in for this data. It should not be read as causal in fact Shapley values can never be read as casual, only and indicator of an impactful predictor.

4.4 Limitations and Future research

In this section the limitations and opportunities for future research is discussed. Regarding the data some major limitation can be pointed out. First, the monthly data ends before the 2020–2024 inflation spike, so the results reflect may or even likely not generalize to the recent high-inflation regime. Also for the evalution period, from 2010 to (and with) 2019, avoids the 2008 financial crisis. This is done since it was observed in the early stages that the transformer models performed even worse for a smaller sample size. Also in the early stages it was discovered that the transformer models preferred a large amount of lags. Furthermore the training implements early stopping which is implemented by taking the last ten steps for an additional validation set that is validated during training. This means that the the transformer models are in practice training on $\frac{9}{10}N - p$, and with a large p (such as 36) this significantly reduces the actual data to train from, thus raising the chance of overfitting (which was also observed in the early stage). From the above analysis there is no reason to believe that transformer models would perform better on more noisy data, thus for the conclusion the exclusion of a more volatile period does not have a huge effect.

Furthermore, as apparent, this thesis is inspired by Medeiros, Schütte, and Soussi (2023). They use the global as well as an regional factor, where the regional factor is the continent average for the continent of the target model. This is not included since, as an attempt to stay as close on the RF baseline created by Medeiros, Schütte, and Soussi (2023) and also let the multi-target models have access to the same amount of input, in an ideal world the models recieve the same amount of inputs. It is recognized that this is not done here as it would differ too much from the RF model in the original. If the interest was more comparing apples to apples, the multivariate RF [37] would be an interesting research topic and would compare more directly to the multi-target models. The multivariate RF was not chosen since it is not nearly as popular in the litterate and becuase Mahalanobis distance is to account for the variance and covariance of multiple outputs when determining the best split points thus making it a different model. But this would make a recursive model using random forest much easier as it would be a single model that made forecasts. In the current setup a recursive method for RF would be an ensemble of all RF models since it needs all forecasts for $h = 1$ in order to forecast $h = 2$, thus also making it an entirely different model. The two linear methods evaluated is both directly comparable to transformer and it is the same model making each recursion.

As mentioned in the introduction the recursive strategy was motivated by the claims of Zeng et al. (2023), and was therefor specifically implemented for the two proposed benchmark. After seeing the performance of especially NLinear in a recursive one wonders how the way more complex transformer models would perform in such a setting, while being highly skeptical of the results it is an option for future research. Another point made by Medeiros, Schütte, and Soussi (2023) is that part of the good performance from RF comes from the nonlinear nature of RF. This could easily be tested by introducing a nonlinearity into NLinear and DLinear. Also with the, compared to the other transformer models, good performance of Informer an interesting angle would be to implement the normalization used by NLinear in Informer.

It is also worth to note that Zhou et al. (2021), Wu et al. (2022), Zhou et al. (2022), evaluate the models on huge horizons (the smallest being 96). It could be that the strength is kept in these huge horizons, and for this comparatively short horizon of 12, there is no notable advantage to the transformer model. The aim here was to have an evaluation of the transformer models to be comparable to the rest of the inflation forecasting literature. Longer horizons are left for future research.

A further limitation is that the analysis relies exclusively on endogenous information, namely past inflation lags and a global factor extracted from these. While this provides a clean and comparable setup across models, it leaves unexplored how these methods perform when richer, more diverse predictors are introduced. Inflation forecasting using the FredMD [30] is often evaluated, while Medeiros et al. (2021) evaluates RF and shows its strength there as well. Transformer models (and the proposed benchmarks) could perform stronger on this dataset, with the ability to handle high-dimensional, heterogeneous inputs even when describing only a single target. Furthermore, there is an enormous amount of transformer models where one claims to beat the other, so there is clearly more to evaluate out there. While this is an interesting angle the three chosen methods have been chosen since these are widely cited and well established in the current forecasting literature.

In addition to the limitations already discussed, an important point concerns the stability of the obtained results. In this thesis the transformer models have only been trained once due to computational constraints, accessibility to a cluster has not been provided thus a limited amount of training could be done [6]. In this rolling window setup Informer and Autoformer was trained and the shap values collected in under 24 hours, while for FEDFormer it took over 48 hours to train and collect shap values for the model with $h = 12$. All of the transformer articles[48, 46, 49], train several models and take the average. While this method would without a doubt make the results stronger, it is both infeasible for this setup due to computational constraints, and one can argue that if a models requires an ensemble of itself to be a strong performer it should be implemented in a better way.

5 Conclusion

The purpose of this thesis was to examine whether transformer-based models can actually improve inflation forecasting compared to strong baselines. Even though transformers have recently received a lot of attention in machine learning and especially natural language processing, little work has tested their usefulness for economic forecasting. To address this, a global panel of monthly inflation data from 91 countries covering 1980–2019 have been used and three transformer variants Informer, Autoformer, and FEDformer have been compared against random forests and DLinear and NLinear, including recursive forecasting using DLinear and NLinear. The forecasts were evaluated across horizons using both error measures (RMSE and MAD) and robustness checks such as the aSPA test.

[6] Google colab has been used to run the transformer models and RF

In answer to the thesis question, transformers do not appear useful for inflation forecasting, as they fail to outperform less complex models in this setting. The strongest performers were recursive purely linear neural networks and random forest, in particular the recursive NLinear, which consistently gave the lowest forecast errors across horizons. DLinear recursive was also highly competitive, confirming that very simple models are still difficult to beat. Random forests was the most consistent performing model, which is in line with [31], especially for the cumulative inflation. While they did not always minimize average errors across horizons, they delivered many of the best forecasts for individual country–horizon pairs and performed well in the average superior predictive ability test against the random walk benchmark. By contrast, only one of the transformers delivered competitive forecasts. Informer performed reasonably well at short horizons, but Autoformer and FEDformer generally lagged behind both the linear and the tree-based benchmarks.

This is result is the classic: adding complexity does not automatically make better forecasts. Transformer models are flexible and powerful, but in practice they require heavy tuning and are computationally costly, and this study shows that they offer little to no improvement in performance for global inflation. Recursive linear neural networks, on the other hand, are straightforward to estimate, run very quickly, and still manage to capture the persistence and seasonality of inflation data. Random forests add value when it comes to picking up local nonlinearities and delivering competitive cumulative forecasts, but they are not dominant for all countries or horizons either. This shows that lightweight models may actually be the most practical option for institutions that need reliable and timely forecasts rather than cutting-edge architectures that are expensive to implement.

This thesis provides one of the first evaluations of transformers for inflation forecasting. A lot of the recent discussion about these models have been based on results from other domains, and this study indicates there are almost no benefits to implement these for macroeconomic forecasting. Furthermore, the results reinforce an important point already emphasized in earlier work: simple models set a very high bar. Even after decades of research and new computational methods, linear methods, recursive forecasting, and random forests is strong compared with more complex deep learning approaches. This shows that progress in forecasting is not only about building more complex architectures, but also about testing carefully whether new methods actually add value in practice.

Lastly, this thesis shows that recursive models and random forests are robust, efficient, and accurate tools for forecasting inflation across a large global dataset, while transformer-based models contribute little beyond added complexity. The results provide a caution against adopting fashionable methods without proper evaluation and highlight the need to balance accuracy, interpretability, and efficiency when choosing forecasting approaches.

A Summary statistics for global inflation data

	mean	std	min	25%	median	75%	max
Austria	0.20	0.39	-1.09	0.00	0.19	0.38	2.41
Burundi	0.76	2.05	-5.04	-0.43	0.52	1.73	15.76
Belgium	0.22	0.31	-0.61	0.01	0.18	0.37	1.61
Burkina Faso	0.24	1.94	-12.44	-0.65	0.17	1.02	11.75
Bahamas	0.25	0.44	-1.73	0.00	0.18	0.40	2.85
Bolivia	-0.00	6.40	-81.68	-0.43	0.06	0.44	51.62
Brazil	-0.01	3.16	-45.67	-0.26	0.02	0.35	10.07
Barbados	0.34	0.81	-4.15	-0.10	0.29	0.71	6.69
Botswana	0.68	0.57	-1.44	0.28	0.60	0.98	3.68
Canada	0.24	0.39	-1.04	0.00	0.22	0.46	2.59
Switzerland	0.13	0.36	-1.05	-0.07	0.11	0.32	1.54
Chile	0.72	0.91	-1.31	0.17	0.47	1.11	7.87
Cote d'Ivoire	0.33	1.27	-5.30	-0.19	0.22	0.68	7.69
Cameroon	0.36	1.35	-7.70	-0.20	0.18	0.68	9.17
Colombia	1.03	0.97	-0.79	0.28	0.79	1.58	4.71
Costa Rica	1.06	1.24	-1.74	0.34	0.85	1.35	10.22
Cyprus	0.25	0.93	-2.52	-0.22	0.36	0.82	3.37
Germany	0.17	0.33	-1.04	0.00	0.12	0.36	1.73
Dominica	0.21	0.94	-4.41	-0.17	0.17	0.57	5.07
Denmark	0.24	0.42	-0.66	-0.10	0.20	0.48	2.54
Dominican Republic	0.95	1.63	-3.33	0.13	0.51	1.22	10.93
Algeria	0.68	1.96	-5.10	-0.40	0.51	1.54	10.79
Ecuador	1.61	1.93	-0.76	0.22	0.98	2.49	13.39
Egypt, Arab Rep.	0.91	1.68	-7.25	0.16	0.65	1.55	9.46
Spain	0.36	0.57	-1.93	0.07	0.31	0.64	2.83
Ethiopia	0.69	2.19	-5.96	-0.47	0.66	1.69	12.00
Finland	0.24	0.40	-0.75	-0.02	0.20	0.42	1.93
Fiji	0.35	0.78	-2.08	-0.10	0.26	0.67	5.38
France	0.23	0.36	-1.01	0.01	0.19	0.40	1.68
United Kingdom	0.28	0.43	-0.70	0.00	0.27	0.44	3.35
Ghana	1.86	2.71	-9.85	0.36	1.23	3.01	21.06
Gambia, The	0.69	1.33	-3.53	0.12	0.43	0.94	11.65
Greece	0.64	1.37	-2.03	-0.30	0.45	1.73	4.53
Grenada	0.25	0.67	-2.68	-0.06	0.14	0.44	4.18

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	mean	std	min	25%	median	75%	max
Guatemala	0.72	1.07	-2.51	0.20	0.50	0.97	7.15
Honduras	0.75	0.78	-1.50	0.29	0.57	0.99	5.23
Haiti	0.90	1.47	-4.62	0.26	0.72	1.43	12.37
Hungary	0.76	1.15	-1.59	0.12	0.53	1.11	9.43
Indonesia	0.71	1.14	-1.07	0.17	0.47	0.93	11.91
India	0.63	0.82	-2.12	0.00	0.64	1.11	4.47
Ireland	0.29	0.82	-1.61	0.00	0.00	0.48	7.13
Iceland	0.89	1.42	-1.47	0.13	0.44	1.06	9.97
Israel	1.68	3.56	-1.49	0.00	0.45	1.31	23.81
Italy	0.35	0.40	-0.58	0.10	0.27	0.46	2.06
Jamaica	1.11	1.28	-1.11	0.42	0.81	1.38	9.73
Jordan	0.35	1.20	-6.62	-0.23	0.28	0.81	7.22
Japan	0.07	0.42	-1.02	-0.21	0.00	0.29	2.07
Kenya	0.90	1.50	-3.12	0.21	0.67	1.35	10.98
St. Kitts and Nevis	0.23	0.71	-2.13	-0.14	0.14	0.44	4.98
Korea, Rep.	0.35	0.57	-1.19	0.00	0.29	0.59	3.65
St. Lucia	0.24	0.92	-2.55	-0.28	0.18	0.74	4.75
Sri Lanka	0.76	1.37	-3.57	0.00	0.75	1.48	5.68
Luxembourg	0.22	0.55	-1.68	0.01	0.17	0.38	1.89
Morocco	0.29	0.80	-2.05	-0.18	0.25	0.73	3.55
Madagascar	1.00	1.72	-9.47	0.23	0.70	1.60	10.59
Mexico	1.59	2.13	-0.74	0.39	0.71	1.95	14.38
Malta	0.19	0.67	-2.28	-0.15	0.24	0.56	3.04
Myanmar	1.20	2.07	-8.37	0.10	0.87	2.31	9.84
Mauritius	0.48	0.83	-2.49	0.08	0.36	0.77	5.27
Malawi	1.45	3.10	-9.25	-0.60	1.29	3.37	13.41
Malaysia	0.23	0.41	-1.51	0.00	0.19	0.38	3.87
Niger	0.21	2.29	-18.42	-0.79	0.08	1.13	14.43
Nigeria	1.38	2.07	-3.57	0.45	0.92	2.03	11.18
Netherlands	0.18	0.43	-1.31	-0.11	0.16	0.46	1.22
Norway	0.29	0.50	-1.28	0.00	0.24	0.53	3.21
Nepal	0.65	1.53	-3.09	-0.15	0.62	1.43	7.95
Pakistan	0.65	0.79	-1.56	0.15	0.56	1.14	4.40
Panama	0.16	0.33	-0.86	0.00	0.10	0.33	1.40
Peru	-0.01	9.32	-147.44	-0.35	0.02	0.34	111.30
Philippines	0.61	0.95	-1.97	0.18	0.39	0.76	8.56
Portugal	0.51	0.81	-1.42	0.01	0.34	0.82	4.51

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	mean	std	min	25%	median	75%	max
Paraguay	0.91	1.40	-6.00	0.10	0.66	1.47	10.05
Saudi Arabia	0.11	0.53	-4.91	-0.10	0.09	0.30	4.18
Sudan	2.59	4.54	-15.22	0.14	1.95	4.37	24.93
Senegal	0.27	1.51	-3.92	-0.60	0.08	1.00	11.88
Singapore	0.15	0.46	-1.62	-0.12	0.11	0.39	1.99
Solomon Islands	0.68	1.27	-3.71	-0.03	0.46	1.24	9.11
El Salvador	0.64	0.94	-3.46	0.05	0.40	1.14	4.57
Suriname	1.93	4.04	-5.89	0.13	0.65	2.24	34.16
Slovenia	2.63	5.69	-2.07	0.17	0.75	2.32	45.86
Sweden	0.26	0.54	-1.35	-0.05	0.22	0.51	2.81
Eswatini	0.74	1.97	-14.38	0.16	0.50	1.06	18.14
Seychelles	0.34	1.85	-18.25	-0.20	0.16	0.53	21.62
Thailand	0.27	0.54	-3.06	0.00	0.22	0.55	3.56
Trinidad and Tobago	0.56	0.82	-2.03	0.02	0.38	0.92	6.32
Turkey	2.52	2.33	-1.45	0.72	1.93	4.08	21.01
Taiwan, China	0.17	0.88	-2.61	-0.34	0.14	0.64	4.42
Uruguay	1.98	2.12	-0.73	0.46	1.11	3.18	14.34
United States	0.25	0.34	-1.93	0.07	0.24	0.44	1.51
Samoa	0.47	1.84	-4.96	-0.58	0.33	1.32	11.95
South Africa	0.68	0.59	-1.14	0.28	0.61	1.06	3.60
Global Factor	0.78	0.53	-0.29	0.36	0.59	1.18	3.10

Table 9: Summary statistics of monthly inflation rates across 91 countries, 1980–2019.

B FEDformer

Algorithm 1: Frequency Enhanced Decomposition (FEB-f) Block

```
Input  : x ∈ ℝ^{N×D}: Input time series
         W ∈ ℝ^{D×D}: Learnable projection matrix
         R ∈ ℂ^{D×D×M}: Learnable frequency kernel
         M ≪ N: Number of selected frequency modes
Output:  output ∈ ℝ^{N×D}: Enhanced time series output
Function FEB_f_block(x,W,R,M):
    q ← xW ;                          // Linear projection, shape (N,D)
    Q ← FFT(q) ;  // Discrete Fourier Transform, shape (N,D)
    Q_selected ← SelectModes(Q,M) ;             // Select top M
     frequencies, shape (M,D)
    Initialize Y ∈ ℂ^{M×D} with zeros;
    for m ← 1 to M do
        for d_out ← 1 to D do
            Y[m,d_out] ← 0;
            for d_in ← 1 to D do
                Y[m,d_out] ← Y[m,d_out] + Q_selected[m,d_in]·R[d_in,d_out,m];
    Y_padded ← ZeroPad(Y,N) ;  // Pad zeros to length N, shape
     (N,D)
    output ← IFFT(Y_padded) ;       // Inverse FFT to time domain
    return output
```

Algorithm 2: Frequency Enhanced Attention with Wavelets (FEA-w) Block

Input : $q, k, v \in \mathbb{R}^{N \times D}$: Input query, key, value
$L_{\max}$: Maximum decomposition level

Output: output $\in \mathbb{R}^{N \times D}$: Attention-enhanced output

Function `FEA_w`(q, k, v, L)**:**

```
if L > L_max then
    return FEB_f_block(q, k, v) ;  // Base case: fallback to frequency attention
// Wavelet decomposition (e.g., Legendre multiwavelets)
s_q, d_q, q_res ← WaveletDecompose(q);
s_k, d_k, k_res ← WaveletDecompose(k);
s_v, d_v, v_res ← WaveletDecompose(v);
// Frequency attention on wavelet components
Y_s ← FEB_f_block(s_q, s_k, s_v) ;                 // Low-frequency
Y_d ← FEB_f_block(d_q, d_k, d_v) ;                 // High-frequency
Y_r ← FEA_w(q_res, k_res, v_res, L + 1) ;          // Residual recursion
// Wavelet reconstruction
output ← WaveletReconstruct(Y_s, Y_d, Y_r);
return output
```

C aSPA

Algorithm 3: Average Superior Predictive Ability (aSPA) Test

Input : Loss differential matrix $D \in \mathbb{R}^{T \times H}$
Weights w_h (default $1/H$)
Block length L, bootstrap replications B

Output: Observed statistic t_{obs}, p-value, $\mu_{ij}^{(\text{Avg})}$, $\hat{\omega}_{ij}$

Function `aSPA_test(D,w,L,B)`**:**
- Compute weighted series $z_t = \sum_h w_h D_{t,h}$;
- Estimate $\mu_{ij}^{(\text{Avg})} = \frac{1}{T} \sum_t z_t$;
- Estimate long-run variance $\hat{\omega}_{ij}$ using QS HAC;
- Compute observed statistic $t_{\text{obs}} = \sqrt{T}\, \mu_{ij}^{(\text{Avg})} / \sqrt{\hat{\omega}_{ij}}$;
- **for** $b \leftarrow 1$ **to** B **do**
  - Generate bootstrap sample $z_t^{(b)}$ using moving block bootstrap;
  - Estimate variance $\hat{\zeta}^{(b)}$ from block sums;
  - Compute bootstrap statistic $t^{(b)} = \sqrt{T}\, \bar{z}^{(b)} / \sqrt{\hat{\zeta}^{(b)}}$;
- Compute p-value = proportion of $t^{(b)} \geq t_{\text{obs}}$;
- **return** t_{obs}, p-value, $\mu_{ij}^{(\text{Avg})}, \hat{\omega}_{ij}$;

D Results including h=1 and h=6

Table 10: Forecast accuracy comparison: RMSE and MAD including models trained for smaller h. Minima are highlighted.

RMSE by Horizon												
Model	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
Autoformer h1	0.860											
Autoformer h6	0.787	0.768	0.758	0.761	0.793	0.797						
Autoformer h12	0.813	0.773	0.764	0.752	0.736	0.723	0.726	$\underline{0.729}$	$\underline{0.726}$	0.741	0.802	0.797
FEDFormer h1	0.930											
FEDFormer h6	0.844	0.824	0.800	0.812	0.840	0.846						
FEDFormer h12	0.874	0.840	0.838	0.807	0.788	0.771	0.768	0.760	0.773	0.801	0.848	0.858
Informer h1	0.759											
Informer h6	0.728	0.728	$\underline{0.722}$	0.732	0.726	0.741						
Informer h12	0.720	0.725	0.738	$\underline{0.719}$	0.730	0.734	$\underline{0.722}$	0.731	0.735	$\underline{0.720}$	0.721	0.737
RF	$\boxed{0.678}$	$\underline{0.723}$	0.733	0.759	0.748	**0.690**	0.738	0.754	0.737	0.722	$\underline{0.691}$	$\boxed{0.574}$
DLinear h1	0.740											
DLinear h6	0.743	0.753	0.741	0.736	0.740	0.739						
DLinear h12	0.746	0.761	0.769	0.766	0.763	0.763	0.765	0.765	0.760	0.758	0.759	0.762
NLinear h1	**0.692**											
NLinear h6	$\underline{0.700}$	$\boxed{0.707}$	**0.705**	**0.696**	**0.694**	$\underline{0.693}$						
NLinear h12	0.722	0.733	0.736	0.734	0.741	0.735	0.731	0.730	0.729	0.727	0.722	0.724
DLinear$_{rec}$	0.736	0.752	0.740	0.724	$\underline{0.705}$	0.696	**0.693**	**0.684**	**0.678**	**0.677**	**0.670**	$\underline{0.664}$
NLinear$_{rec}$	$\underline{0.700}$	**0.716**	$\boxed{0.703}$	$\boxed{0.685}$	$\boxed{0.665}$	$\boxed{0.658}$	$\boxed{0.656}$	$\boxed{0.648}$	$\boxed{0.643}$	$\boxed{0.640}$	$\boxed{0.634}$	**0.630**
RW	1.660	1.760	1.796	1.817	1.808	1.770	1.844	1.873	1.842	1.845	1.794	1.657

MAD by Horizon

Model	h=1	h=2	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
Autoformer h1	0.511											
Autoformer h6	0.462	0.450	0.444	0.443	0.463	0.454						
Autoformer h12	0.482	0.451	0.449	0.435	0.423	0.416	0.419	0.421	0.414	0.430	0.474	0.474
FEDFormer h1	0.549											
FEDFormer h6	0.509	0.482	0.477	0.475	0.496	0.500						
FEDFormer h12	0.529	0.501	0.487	0.482	0.469	0.450	0.454	0.450	0.450	0.468	0.505	0.507
Informer h1	0.419											
Informer h6	$\underline{0.400}$	**0.390**	$\boxed{0.404}$	**0.395**	**0.396**	0.405						
Informer h12	**0.398**	$\underline{0.398}$	**0.408**	$\boxed{0.387}$	$\underline{0.399}$	$\underline{0.396}$	**0.395**	**0.397**	$\underline{0.397}$	0.392	0.391	0.404
RF	$\boxed{0.357}$	$\boxed{0.379}$	0.416	0.421	0.415	**0.393**	0.420	0.420	0.415	**0.384**	**0.360**	$\boxed{0.288}$
DLinear h1	0.437											
DLinear h6	0.447	0.454	0.448	0.446	0.446	0.437						
DLinear h12	0.444	0.461	0.455	0.463	0.458	0.468	0.460	0.461	0.449	0.448	0.454	0.453
NLinear h1	0.404											
NLinear h6	0.411	0.424	0.418	0.418	0.408	0.405						
NLinear h12	0.425	0.431	0.446	0.445	0.450	0.447	0.438	0.446	0.433	0.433	0.435	0.431
DLinear$_{r}ec$	0.456	0.461	0.438	0.430	0.414	0.409	$\underline{0.408}$	$\underline{0.400}$	**0.393**	$\underline{0.388}$	$\underline{0.384}$	$\underline{0.372}$
NLinear$_{r}ec$	0.408	0.420	$\underline{0.414}$	$\underline{0.396}$	$\boxed{0.385}$	$\boxed{0.374}$	$\boxed{0.372}$	$\boxed{0.368}$	$\boxed{0.367}$	$\boxed{0.359}$	$\boxed{0.356}$	**0.353**
RW	0.578	0.660	0.695	0.718	0.708	0.684	0.711	0.730	0.719	0.697	0.645	0.533

The table reports average forecast errors across all countries for horizons $h = 1, \ldots, 12$. Performance is measured by Root Mean Squared Error (RMSE) and Median Absolute Deviation (MAD). Boxed values are minima across the horizon, bold values are second smallest and underscored values are third smallest.

Summary Statistics				
Model	**Freq_RMSE**	**Freq_MAD**	**aSPA_mean**	**aSPA_share**
Autoformer h1	0.000	0.001	0.421	0.344
Autoformer h6	0.012	0.016	0.183	<u>0.744</u>
Autoformer h12	0.053	0.048	<u>0.152</u>	**0.767**
FEDFormer h1	0.000	0.001	0.477	0.367
FEDFormer h6	0.002	0.007	0.235	0.622
FEDFormer h12	0.018	0.028	0.214	0.644
Informer h1	0.001	0.000	0.265	0.567
Informer h6	0.038	0.045	0.173	0.700
Informer h12	<u>0.075</u>	<u>0.097</u>	0.174	0.733
RF	$\boxed{0.345}$	$\boxed{0.377}$	$\boxed{0.044}$	$\boxed{0.921}$
DLinear h1	0.000	0.000	0.395	0.456
DLinear h6	0.006	0.009	0.270	0.567
DLinear h12	0.027	0.023	0.303	0.533
NLinear h1	0.010	0.003	0.314	0.533
NLinear h6	0.048	0.028	0.194	0.656
NLinear h12	0.068	0.051	0.231	0.611
DLinear_rec	0.047	0.073	0.192	0.733
NLinear_rec	**0.250**	**0.187**	**0.145**	**0.767**
RW	0.001	0.006		

Table 11: Summary statistics across models.

Notes: Freq_RMSE and Freq_MAD report the share of country–horizon pairs where a model achieves the lowest error. aSPA_mean is the mean p-value of the aSPA test against the random walk, where smaller values indicate stronger evidence against the benchmark. aSPA_share is the fraction of countries for which the null of equal predictive ability is rejected at the 5% level. The best, second-best, and third-best values in each column are boxed, bolded, and underlined, respectively.

E Best performing models per country per horizon

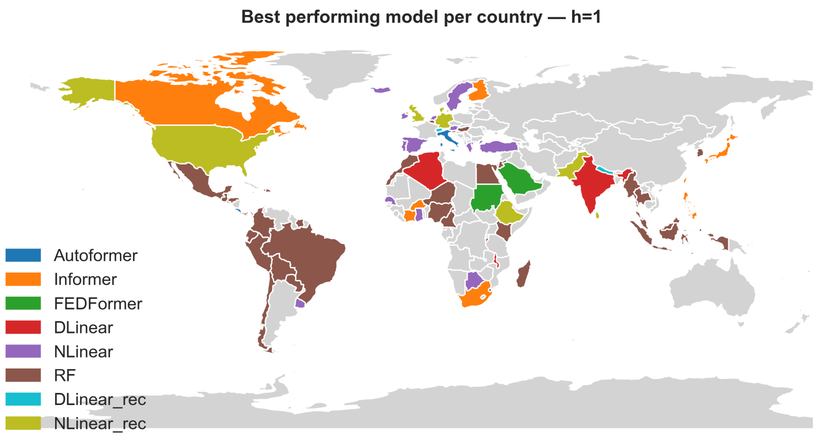

Figure 17: Best-performing forecasting model by country at horizon $h = 1$.

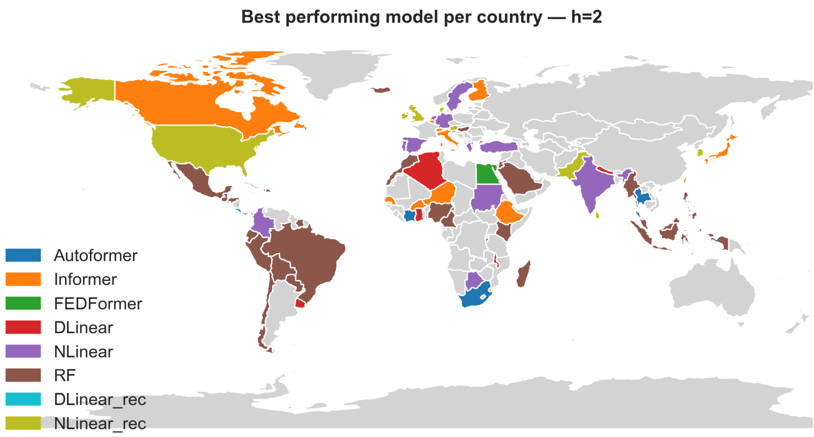

Figure 18: Best-performing forecasting model by country at horizon $h = 2$.

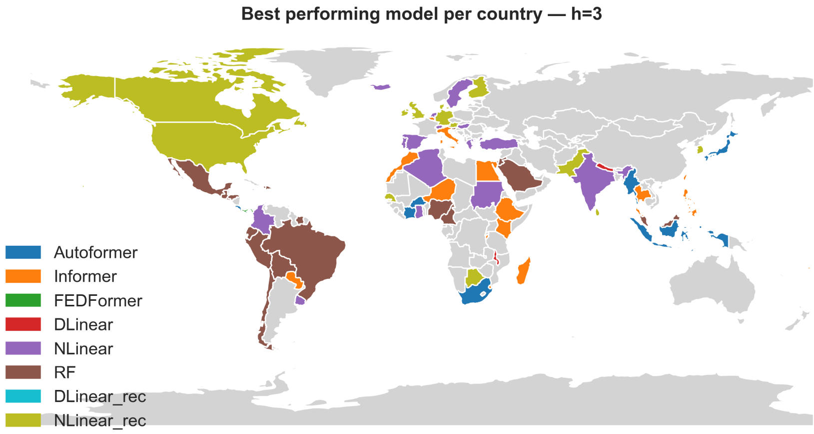

Figure 19: Best-performing forecasting model by country at horizon $h = 3$.

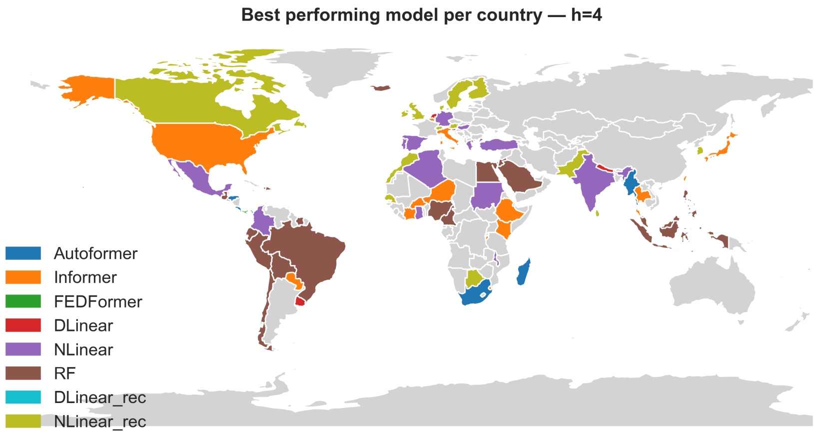

Figure 20: Best-performing forecasting model by country at horizon $h = 4$.

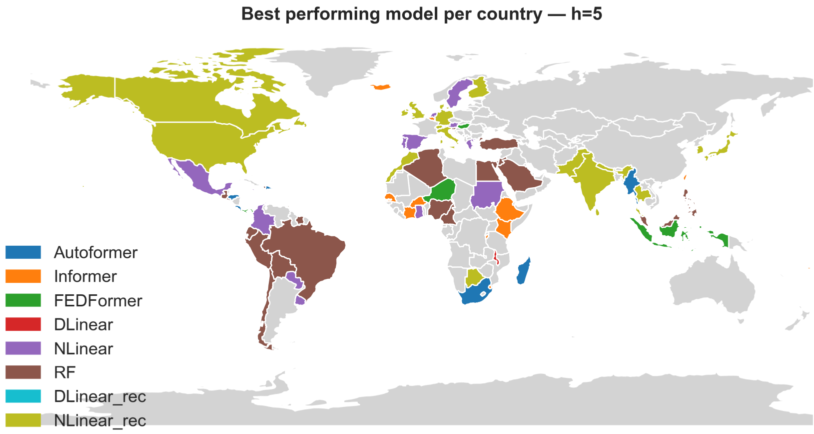

Figure 21: Best-performing forecasting model by country at horizon $h = 5$.

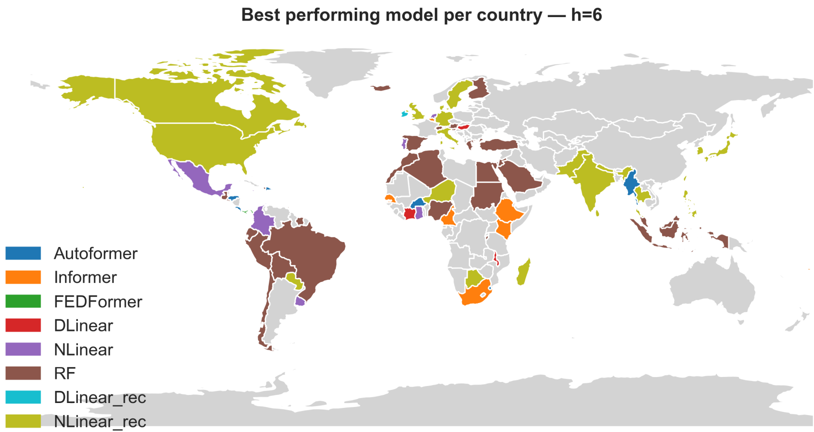

Figure 22: Best-performing forecasting model by country at horizon $h = 6$.

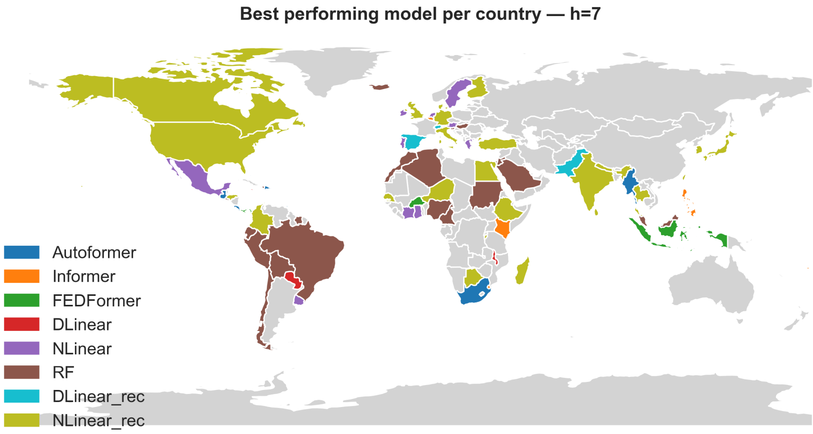

Figure 23: Best-performing forecasting model by country at horizon $h = 7$.

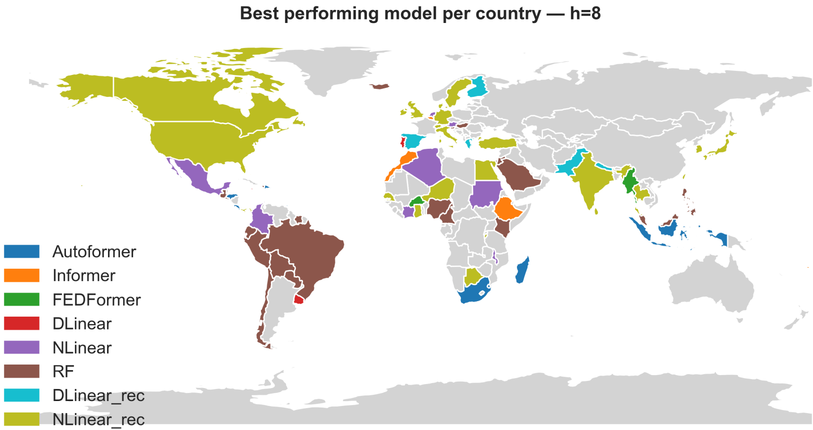

Figure 24: Best-performing forecasting model by country at horizon $h = 8$.

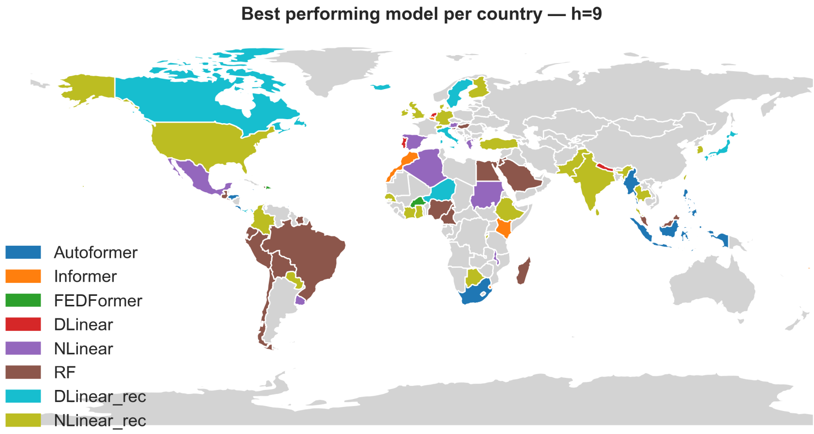

Figure 25: Best-performing forecasting model by country at horizon $h = 9$.

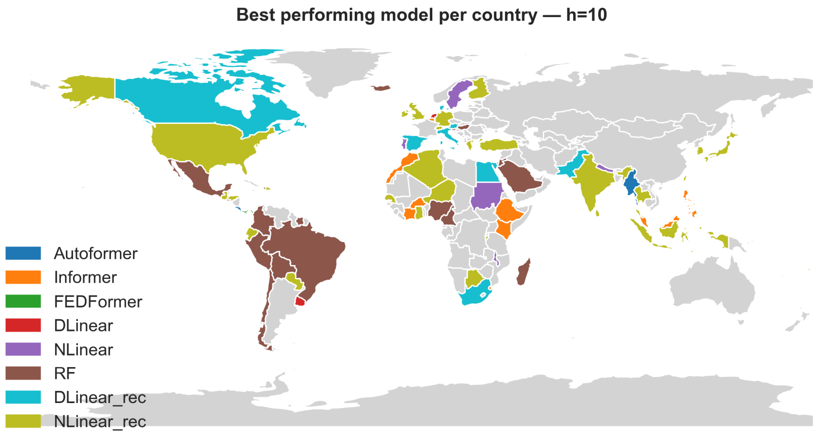

Figure 26: Best-performing forecasting model by country at horizon $h = 10$.

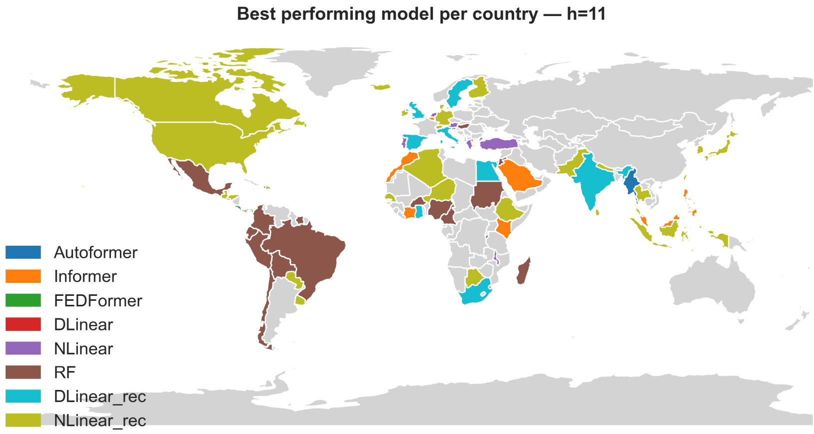

Figure 27: Best-performing forecasting model by country at horizon $h = 11$.

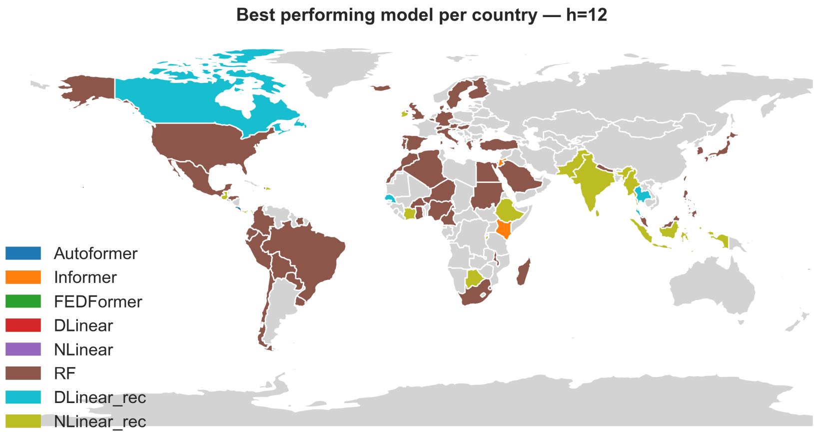

Figure 28: Best-performing forecasting model by country at horizon $h = 12$.

F Shap plots for FEDFormer, DLinear and DLinear recursive

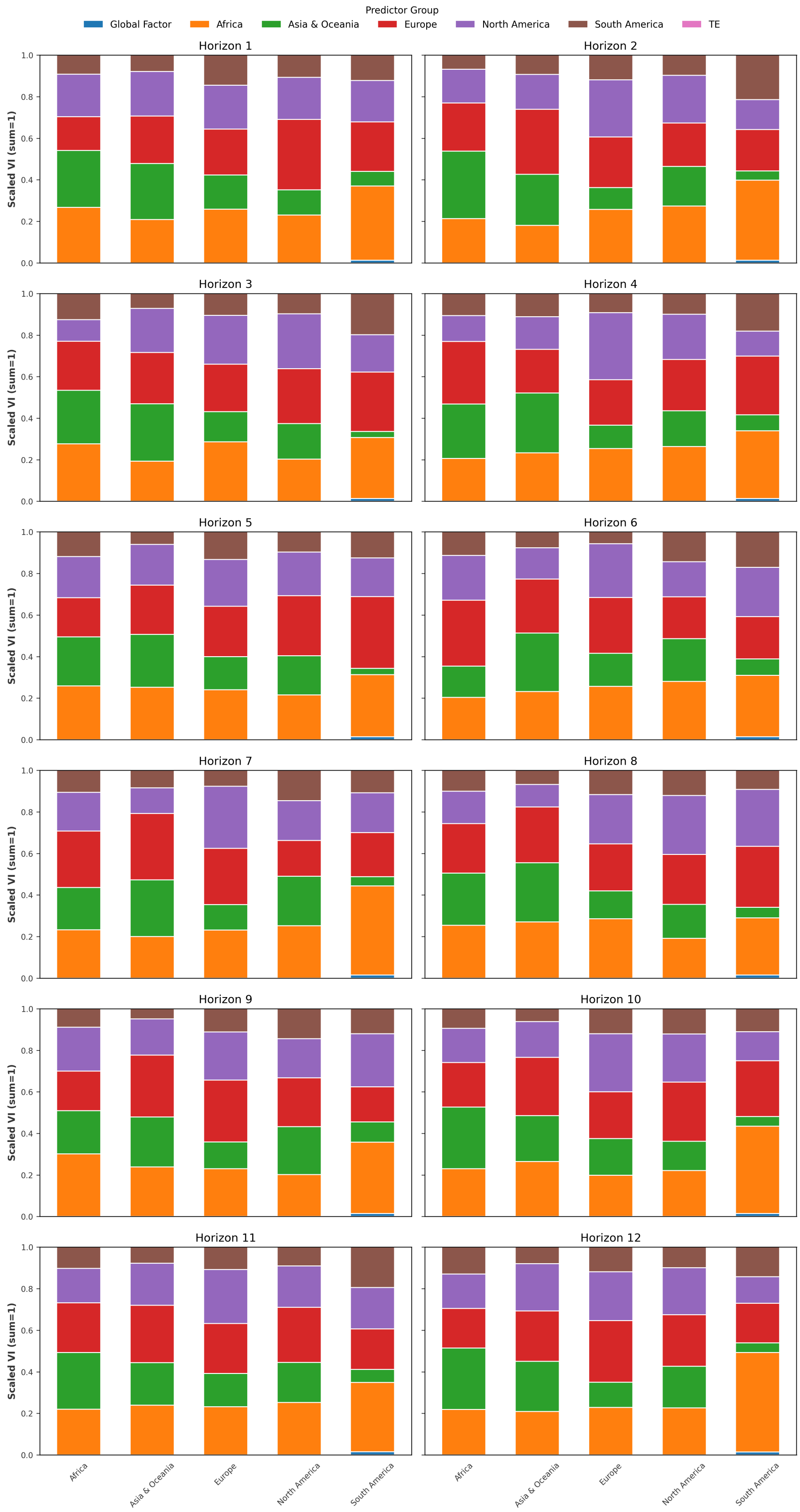

Figure 29: FEDFormer: Predictor groups across horizons and continents.

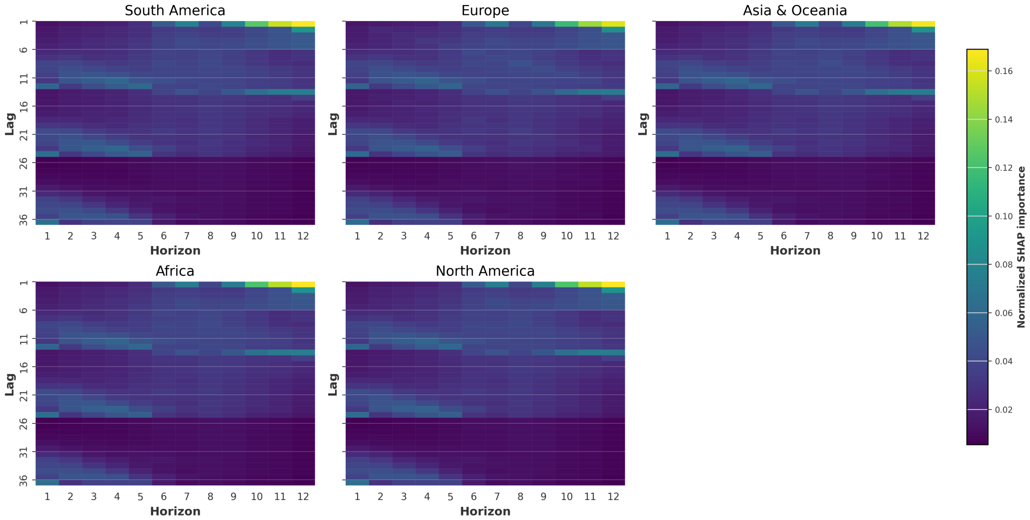

Figure 30: FEDFormer: Lag–horizon heatmap of variable importance.

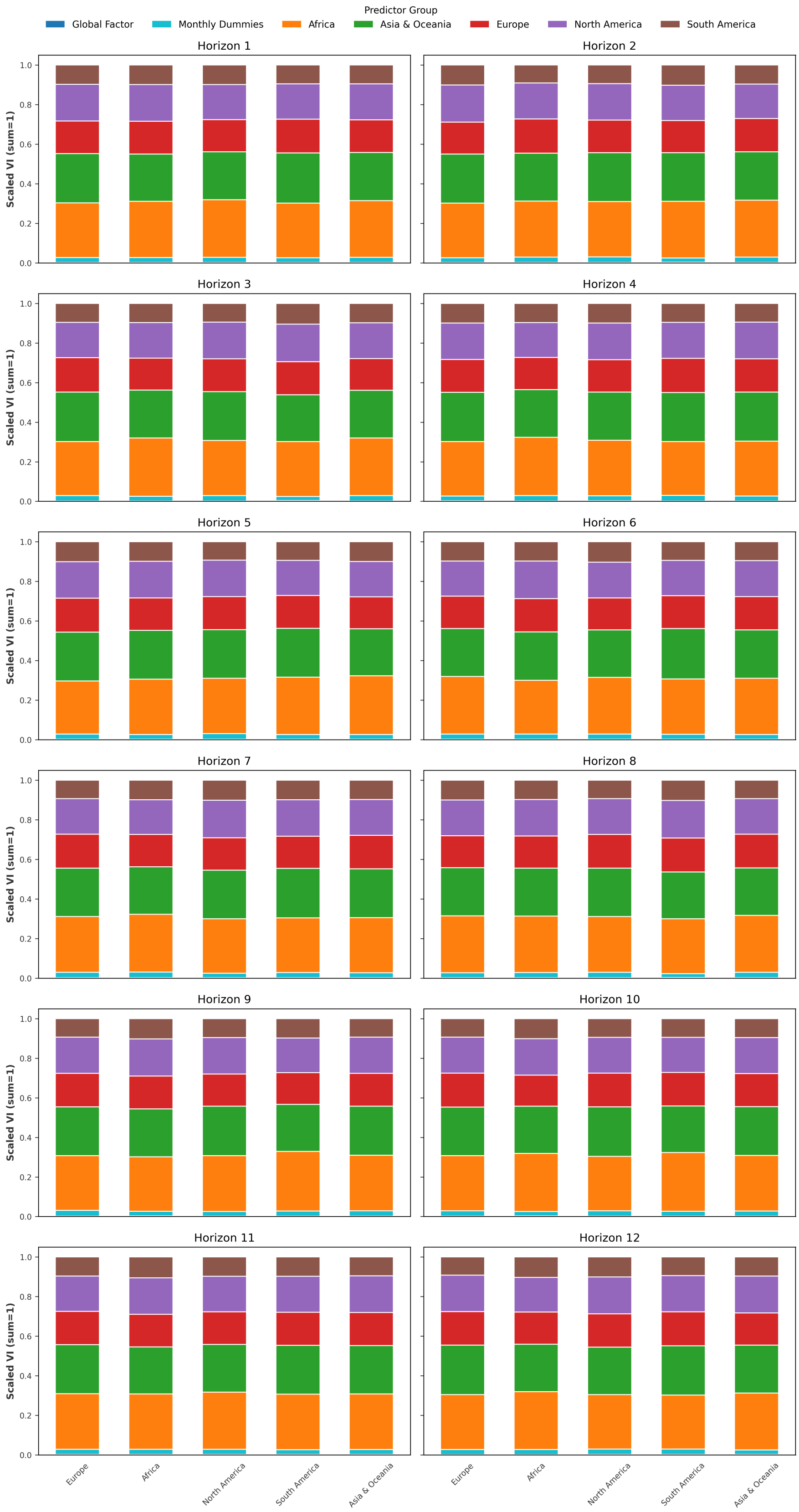

Figure 31: NLinear: Predictor groups across horizons and continents.

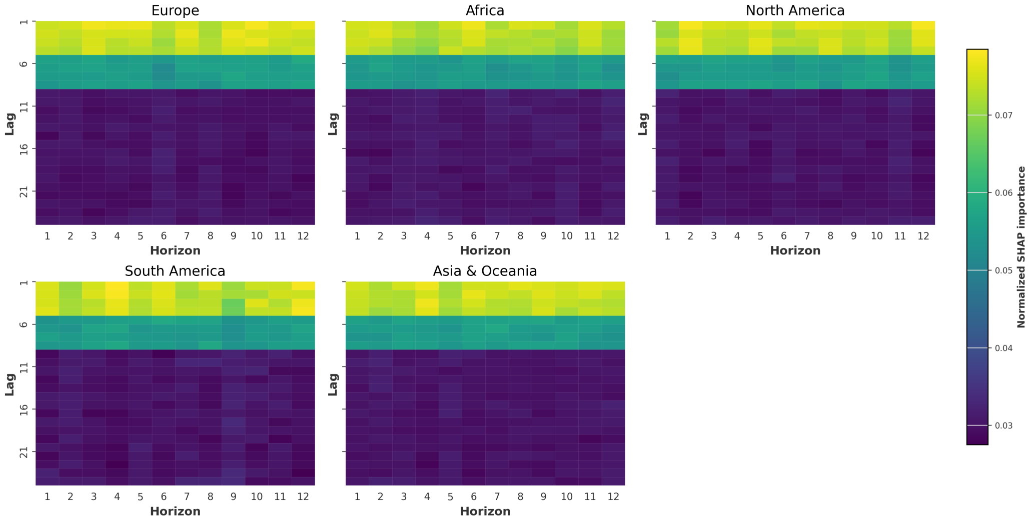

Figure 32: NLinear: Lag–horizon heatmap of variable importance.

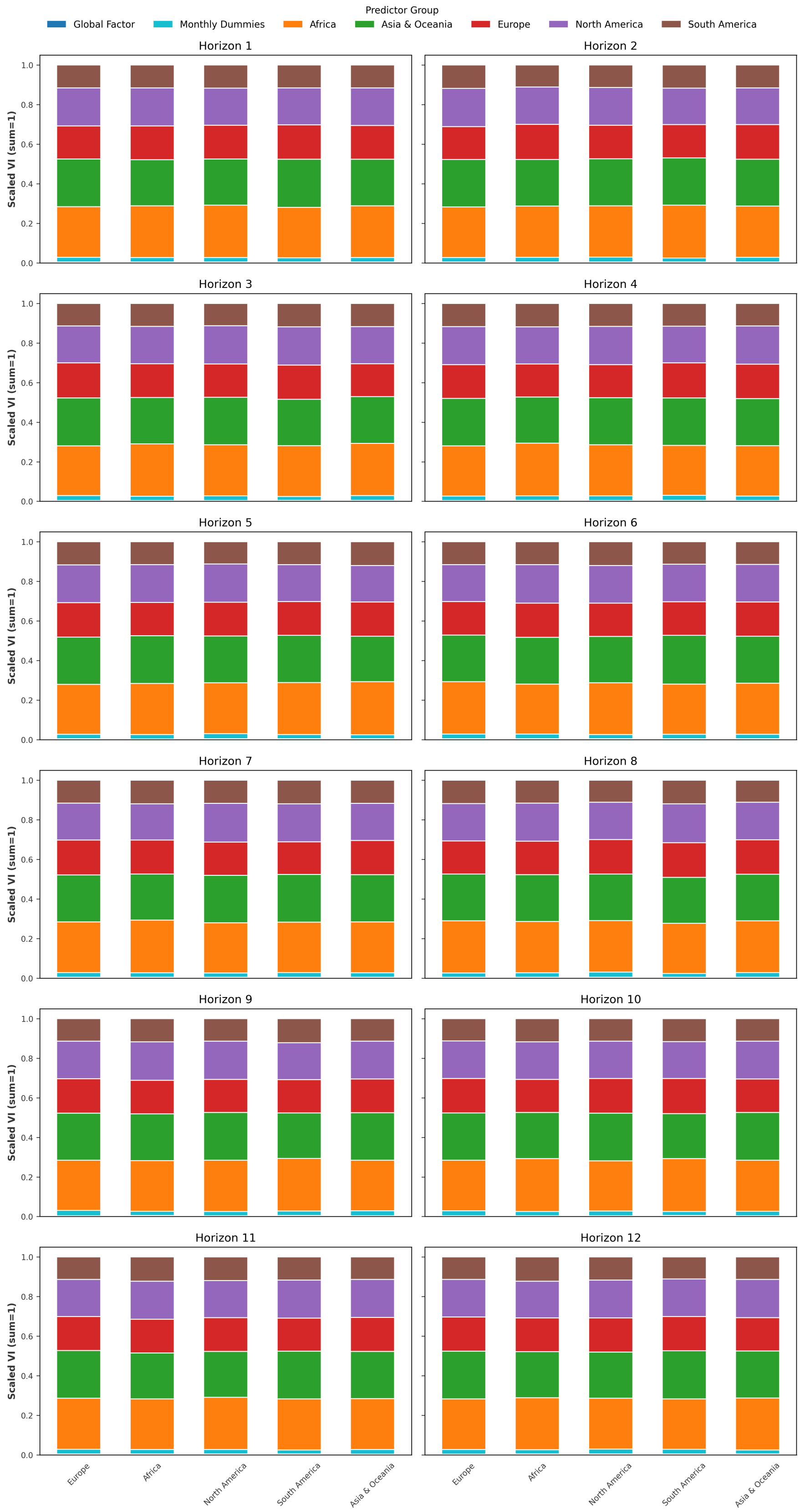

Figure 33: DLinear: Predictor groups across horizons and continents.

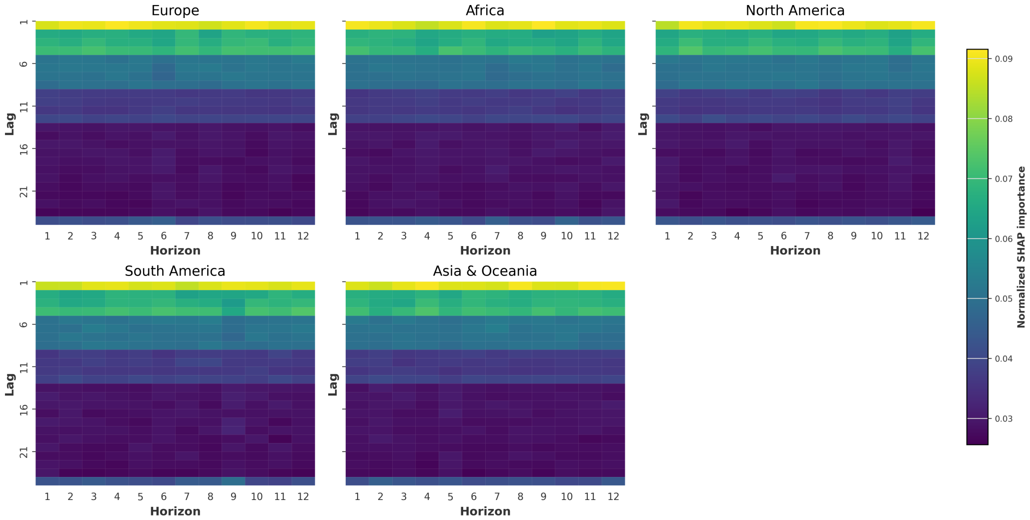

Figure 34: DLinear: Lag–horizon heatmap of variable importance.

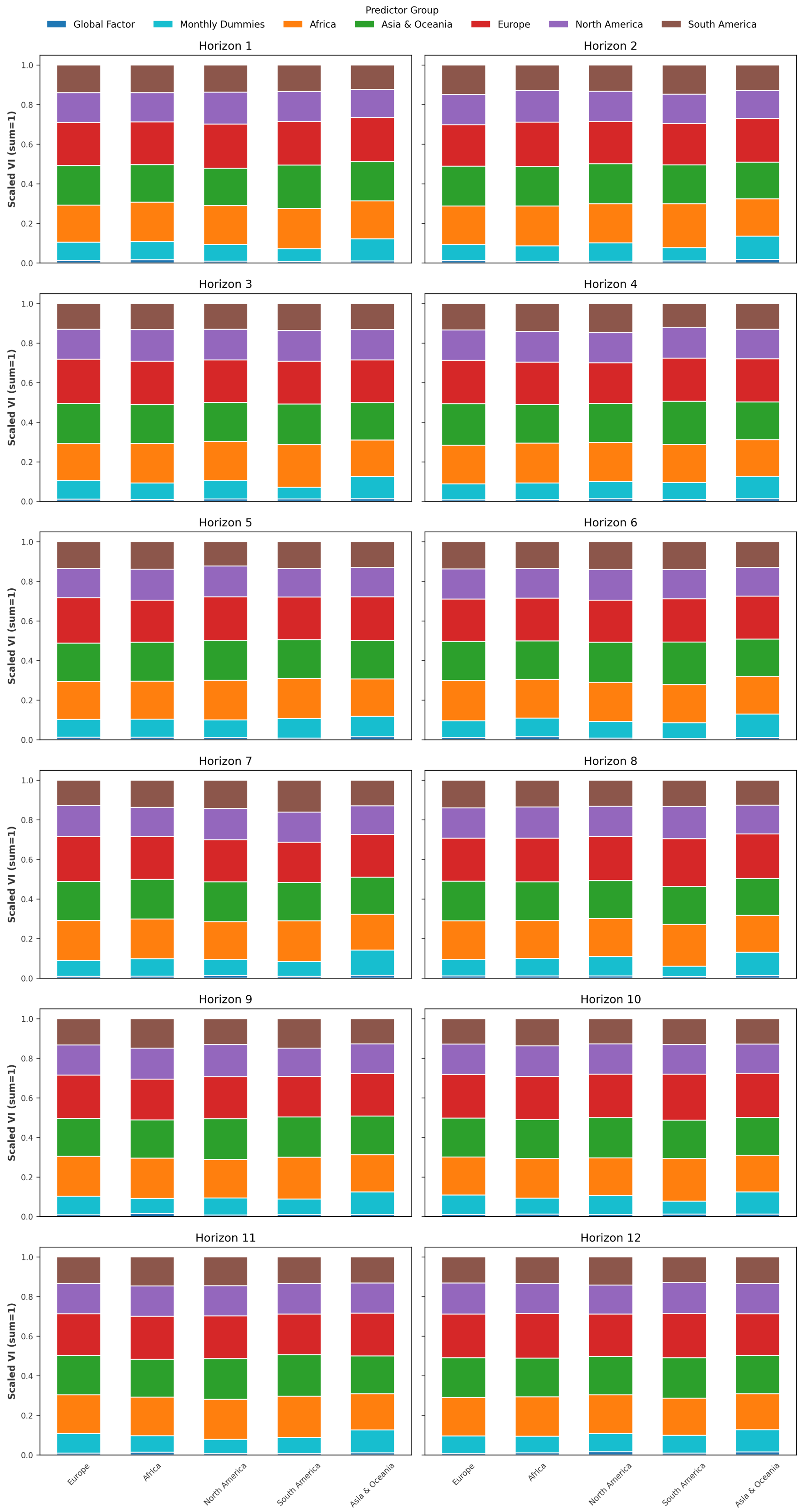

Figure 35: Recursive DLinear: Predictor groups across horizons and continents.

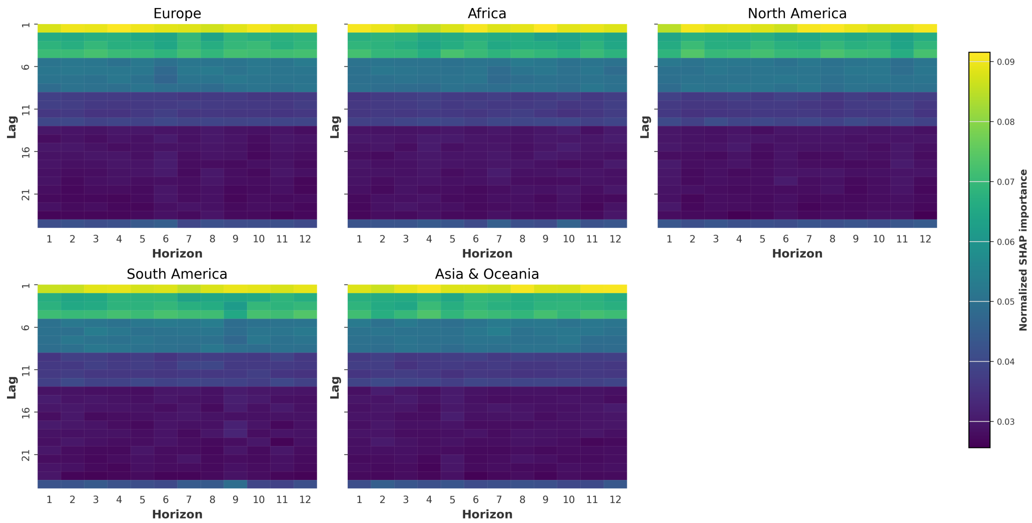

Figure 36: Recursive DLinear: Lag–horizon heatmap of variable importance.

G Note on GAI

Throughout the thesis, GAI was used as a research assistant to discuss methodological choices, debug code, clarify and search for relevant literature. And helped with the formulation of the text.

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